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COLLABORATIVE ECOLOGICAL & BIODIVERSITY RESEARCH

IonCure Tech. Pvt. Ltd.

Himalayan Biodiversity, Fauna & Ecosystem Intelligence Initiative

STATE-WIDE 13-DISTRICT ECOLOGICAL RESEARCH MONOGRAPH

Research Done on Fauna, Wildlife and Ecosystem of Uttarakhand with the Help of IonCure Tech. Pvt. Ltd.

Exhaustive District-by-District Wildlife Catalogue, Protected Area Boundaries, Altitudinal Zonation, Faunal Inventories (Mammals, Avifauna, Herpetofauna, Lepidoptera & Ichthyofauna), Conservation Status & Human-Wildlife Conflict Analysis Across All 13 Districts of Uttarakhand



Bioclimatic Gradient of Uttarakhand (190 m – 7,816 m): Terai-Bhabar Sal Forests, Shivalik Foothills, Temperate Banj Oak-Deodar Ridges, Sub-Alpine Birch Forests, Alpine Bugyals & Nanda Devi Glaciated Peaks



Bengal Tiger (Panthera tigris)



Snow Leopard (Panthera uncia)



State Animal: Alpine Musk Deer



State Bird: Himalayan Monal

13 Districts
Complete Coverage: 6 Kumaon & 7 Garhwal Districts

6 National Parks
7 Wildlife Sanctuaries, 4 Conservation Reserves & 1 Biosphere

38,000 km²
Recorded Forest Area (71.05% of State Geographic Area)

560 Tigers
124 Snow Leopards, 2,026 Elephants & ~710 Bird Species

Research Attribution: Research done on Fauna, Wildlife and Ecosystem of Uttarakhand with the help of **IonCure Tech. Pvt. Ltd.**

Coverage: Almora · Bageshwar · Chamoli · Champawat · Dehradun · Haridwar · Nainital · Pauri · Pithoragarh · Rudraprayag · Tehri · U.S. Nagar · Uttarkashi

Wildlife, Fauna, and Ecosystems of Uttarakhand: Complete 13-District Guide & Catalogue

Spanning 53,483 sq km across the central sector of the Indian Himalayan Region (28°43'N to 31°28'N latitude and 77°34'E to 81°02'E longitude), Uttarakhand features one of the steepest bioclimatic and altitudinal gradients on Earth (Forest Survey of India ISFR). Rising from 190 meters in the southern alluvial plains of the Terai Arc Landscape to 7,816 meters at the summit of Nanda Devi, this vertical relief compresses tropical moist deciduous Sal forests, subtropical Chir pine belts, temperate oak-rhododendron and coniferous canopies, sub-alpine birch-juniper krummholz, alpine meadows (*Bugyals*), and cold-arid Trans-Himalayan steppes into less than 180 kilometers of horizontal width (Uttarakhand Biodiversity Board).

Positioned at the biogeographical crossroads where Indomalayan lowland fauna interdigitates with Palearctic montane and Tibetan plateau species, Uttarakhand encompasses **6 National Parks**, **7 Wildlife Sanctuaries**, **4 Conservation Reserves**, **1 UNESCO Biosphere Reserve** (Nanda Devi and Valley of Flowers National Parks), **2 Tiger Reserves** (Jim Corbett National Park and Rajaji National Park), **1 Elephant Reserve** (Shivalik Elephant Reserve), and **1 Ramsar Wetland of International Importance** (Asan Barrage).

State-Wide Biodiversity & Forest At-a-Glance

- **Recorded Forest Area (RFA):** 38,000 sq km (71.05% of geographical area), managed across 45 Forest and Wildlife Divisions (ISFR Volume II).
- **Satellite-Verified Canopy Forest Cover:** 24,305.13 sq km (45.44% of geographical area), comprising 5,055.01 sq km of Very Dense Forest (canopy ≥70%), 12,768.05 sq km of Moderately Dense Forest (40–70%), and 6,482.07 sq km of Open Forest (10–40%) across 9 Forest Type Groups and 43 forest types (ISFR District Analysis).
- **Faunal Richness:** Over 102 mammal species (ZSI Mammal Checklist), 710+ bird species (over 50% of India's avian diversity), 72 reptile species, 20 amphibian species, 124 freshwater fish species, and 500+ butterfly species (Zoological Survey of India, Wildlife Institute of India).
- **Flagship Predator Populations:** 560 Bengal tigers (3rd highest in India; 260 in Corbett alone, the highest density in any single reserve globally) (All India Tiger Estimation 2022, NTCA Tiger Status Report), 652 Indian leopards in the sampled tiger-forested belt (NTCA Leopard Status 2022, Drishti IAS Analysis), and 124 Snow leopards (2nd highest in India after Ladakh) (SPAI Assessment).
- **Official State Biodiversity Symbols:** State Animal: Alpine musk deer (*Moschus chrysogaster*); State Bird: Himalayan monal (*Lophophorus impejanus*); State Fish: Golden mahseer (*Tor putitora*); State Butterfly: Common Peacock (*Papilio bianor polyctor*); State Tree: Burans (*Rhododendron arboreum*); State Flower: Brahma Kamal (*Saussurea obvallata*) (WII State Symbols).

Official State Biodiversity Symbols of Uttarakhand:



State Animal: Alpine Musk Deer (Moschus chrysogaster) — Kedarnath & Askot WLS



State Bird: Himalayan Monal (Lophophorus impejanus) — High-altitude Bugyals



State Fish: Golden Mahseer (Tor putitora) — Kali, Saryu, Ramganga & Ganges

Complete Directory of Protected Areas in Uttarakhand

Protected Area / Reserve	Statutory Category	Primary District(s)	Year Est.	Area (sq km)	Key Ecological Zone & Flagship Guild
Jim Corbett National Park	National Park & Tiger Reserve	Nainital, Pauri Garhwal	1936 (TR 1973)	520.82 (Core) / 1,288.31 (TR)	Bhabar-Shivalik Sal & Chaurs; Bengal tiger, Indian elephant, Gharial
Rajaji National Park	National Park & Tiger Reserve	Haridwar, Dehradun, Pauri Garhwal	1983 (TR 2015)	820.42 (Core) / 1,075.17 (TR)	Shivalik-Doon Sal & Riverine; Indian elephant, Bengal tiger, King cobra

Protected Area / Reserve	Statutory Category	Primary District(s)	Year Est.	Area (sq km)	Key Ecological Zone & Flagship Guild
Nanda Devi National Park	National Park & UNESCO WHS	Chamoli (bordering Pithoragarh, Bageshwar)	1982 (WHS 1988)	624.60	Inner Glacial Sanctuary & Alpine; Snow leopard, Bharal, Alpine musk deer
Valley of Flowers National Park	National Park & UNESCO WHS	Chamoli	1982 (WHS 2005)	87.50	Zanskar-Great Himalaya Transition Alpine; Alpine musk deer, Bharal, Apollo butterflies
Gangotri National Park	National Park	Uttarkashi	1989	2,390.02	Upper Bhagirathi & Nelang Trans-Himalaya; Snow leopard, Argali, Tibetan sandgrouse
Govind Pashu Vihar National Park	National Park & Sanctuary	Uttarkashi	1955 (WLS) / 1990 (NP)	472.08 (NP) / 957.97 (Total)	Tons-Supin Conifer & Alpine; Snow leopard, Western tragopan, Bearded vulture
Kedarnath Wildlife Sanctuary	Wildlife Sanctuary	Chamoli, Rudraprayag	1972	975.20	Mandakini-Alaknanda Sub-Alpine & Bugyals; Alpine musk deer, Himalayan tahr, Himalayan monal
Askot Musk Deer Sanctuary	Wildlife Sanctuary	Pithoragarh	1986	599.93	Kali-Gori-Dhauli Trans-Boundary Alpine; Alpine musk deer, Snow leopard, Himalayan serow
Sonanadi Wildlife Sanctuary	Wildlife Sanctuary (Corbett Buffer)	Pauri Garhwal	1987	301.18	Sonanadi-Palain Bhabar Deciduous; Bengal tiger, Indian elephant, Golden mahseer
Nandhaur Wildlife Sanctuary	Wildlife Sanctuary	Nainital, Champawat	2012	269.96	Gola-Sharda Terai-Bhabar Corridor; Bengal tiger, Indian elephant, Great hornbill
Binsar Wildlife Sanctuary	Wildlife Sanctuary	Almora	1988	45.59	Jhandi Dhar Temperate Oak-Rhododendron; Indian leopard, Himalayan goral, Koklass pheasant
Mussoorie Wildlife Sanctuary	Wildlife Sanctuary (Benog)	Dehradun	1993	10.82	Aglar Catchment Oak-Pine; Cheer pheasant, Mountain quail (historical), Himalayan goral
Naina Devi Himalayan Bird Conservation Reserve	Conservation Reserve	Nainital	2015	111.91	Pangot-Kilbury-Vinayak Temperate Oak; >250 Himalayan bird species, Himalayan serow
Pawalgarh Conservation Reserve	Conservation Reserve	Nainital	2012	58.25	Dabka-Kosi Bhabar Corridor; Bengal tiger, Indian elephant, Ibisbill
Jhilmil Jheel Conservation Reserve	Conservation Reserve	Haridwar	2005	37.84	Gangetic Terai Oxbow Swamp; Swamp deer (Barasingha), Indian elephant, Burmese python
Asan Barrage (Asan Conservation Reserve)	Conservation Reserve & Ramsar Site	Dehradun	2005 (Ramsar 2020)	4.44	Asan-Yamuna Confluence Wetland; Red-headed vulture, Baer's pochard, Golden mahseer

District-Wise Forest Cover & Division Matrix (ISFR Assessment)

District	Division	Geographical Area (sq km)	Very Dense Forest (sq km)	Mod. Dense Forest (sq km)	Open Forest (sq km)	Total Forest Cover (sq km)	% of District Area	Key Territorial & Wildlife Forest Divisions
Almora	Kumaon	3,144	199.09	994.16	526.55	1,719.80	54.70%	Almora FD, Civil Soyam Almora, Binsar WLS
Bageshwar	Kumaon	2,241	162.39	761.61	338.69	1,262.69	56.34%	Bageshwar FD, Nanda Devi Biosphere Buffer
Chamoli	Garhwal	8,030	443.00	1,580.68	686.43	2,710.11	33.75%	Badrinath FD, Kedarnath WLS FD, Nanda Devi NP FD
Champawat	Kumaon	1,766	339.09	578.96	305.95	1,224.00	69.32%	Champawat FD, Haldwani FD (Nandhaur WLS East)

District	Division	Geographical Area (sq km)	Very Dense Forest (sq km)	Mod. Dense Forest (sq km)	Open Forest (sq km)	Total Forest Cover (sq km)	% of District Area	Key Territorial & Wildlife Forest Divisions
Dehradun	Garhwal	3,088	663.25	636.33	312.00	1,611.58	52.19%	Dehradun FD, Chakrata FD, Mussoorie FD, Kalsi FD
Haridwar	Garhwal	2,360	99.00	315.00	174.00	588.00	24.74%	Haridwar FD, Rajaji TR (Haridwar & Chilla Ranges)
Nainital	Kumaon	4,251	762.49	1,690.00	592.00	3,044.49	71.62%	Nainital FD, Ramnagar FD, Haldwani FD, Terai Central/East/West
Pauri Garhwal	Garhwal	5,329	620.71	1,920.00	856.00	3,396.71	63.74%	Garhwal FD, Lansdowne FD, Kalagarh TR FD, Civil Soyam Pauri
Pithoragarh	Kumaon	7,090	517.60	975.47	584.93	2,078.00	29.35%	Pithoragarh FD, Askot WLS Range (Dharchula/Munsiyari)
Rudraprayag	Garhwal	1,984	246.30	588.00	308.00	1,142.30	57.58%	Rudraprayag FD, Kedarnath WLS FD (Guptkashi/Ukhimath)
Tehri Garhwal	Garhwal	3,642	356.39	1,180.00	528.00	2,064.39	56.68%	Tehri FD, Narendra Nagar FD, Tehri Dam I & II FD
Udham Singh Nagar	Kumaon	2,542	121.00	205.00	102.00	428.00	16.84%	Terai East FD (Surai/Khatima), Terai Central FD, Terai West FD
Uttarkashi	Garhwal	8,016	566.15	1,680.00	790.00	3,036.15	37.88%	Uttarkashi FD, Tons FD, Upper Yamuna Barkot FD, Gangotri NP, Govind NP
State Total	**13 Districts**	**53,483**	**5,055.01**	**12,768.05**	**6,482.07**	**24,305.13**	**45.44%**	**45 Forest & Wildlife Divisions Across 3 Regions**

1. Almora District (Kumaon Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Almora district (3,144 sq km; official portal almora.nic.in) anchors the central Lesser Himalayan ridge system of Kumaon. Its primary statutory protected area is Binsar Wildlife Sanctuary, notified in 1988 across 45.59 sq km on the Jhandi Dhar mountain spur roughly 30 km north of Almora town. Spanning an elevation range of 900 to 2,500 meters (with Zero Point at 2,412 m), Binsar is surrounded by an Eco-Sensitive Zone encompassing Regariya and adjacent Van Panchayat community forests. Additionally, the western flank of Almora (Marchula-Mohaan sector in Salt and Bhikiyasain blocks) directly abuts the northern buffer corridor of Jim Corbett National Park, creating a continuous vertical dispersal pathway from the 450-meter Ramganga valley up to 2,750-meter temperate ridges.
- **Forest Cover & Types:** Managed by the Almora Forest Division and Civil Soyam Forest Division Almora, the district maintains 1,719.80 sq km of forest cover (54.70% of geographical area), comprising 199.09 sq km of Very Dense Forest, 994.16 sq km of Moderately Dense Forest, and 526.55 sq km of Open Forest (ISFR District Data). Between 800 and 1,800 meters, Himalayan Sub-tropical Pine Forests (Champion & Seth Type 9/C1b) dominated by Chir pine (*Pinus roxburghii*) occupy sun-exposed southern slopes. Above 1,800 meters across Binsar, Jageshwar, Ranikhet, Sitlakhet, and Bhatrojkhan, Moist Temperate Broadleaf Forests (Type 12/C1a) prevail, dominated by Banj oak (*Quercus leucotrichophora*), Rianj oak (*Quercus lanata*), Tilonj oak (*Quercus floribunda*), and ruby-flowered Burans (*Rhododendron arboreum*) (Quercus lanata Spatial Study). At Jageshwar Dham (1,870 m) and Pandavkholi (2,600 m), sacred groves of ancient Himalayan Deodar (*Cedrus deodara*) and cypress (*Cupressus torulosa*) open into high montane grassy glades (*Thatch* or low-elevation bugyals) around Dunagiri, Bhatkot, and Mornaula.
- **Aquatic Ecosystems & Water Bodies:** Drained by two non-glacial, spring-fed perennial river basins: the **Kosi River** (originating at Dharpani Dhar near the Kausani border and flowing through Someshwar and Hawalbagh valleys) and the **Western Ramganga River** (coursing through Chaukhutiya, Bhikiyasain, and Marchula). Key tributaries include the

Suyal River (joining the Kosi at Quarab), **Gagas River** (draining Dwarahat and Ranikhet), **Nana Kosi**, and **Bino River**. Important lentic water bodies and wetlands include **Bhalu Dam** (constructed in 1903 near Chaubatia, Ranikhet) and **Tadagtal** (a natural tectonic lake bed with five subterranean outlets near Chaukhutiya).

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 1. Almora District:



Mammal: Indian Leopard (Panthera pardus fusca) — Binsar Wildlife Sanctuary



Bird: Himalayan Monal (Lophophorus impejanus) — High ridges of Pandavkholi & Dunagiri



Reptile: King Cobra (Ophiophagus hannah) — Western Ramganga & Marchula foothills



Mammal: Himalayan Serow (Capricornis sumatraensis thar) — Binsar & Jageshwar oak-ringal gorges

- **Mammals:** The Indian leopard (*Panthera pardus fusca*) is the primary apex predator across all 11 tehsils from 600 to 2,600 meters, while camera-trap monitoring in Binsar Wildlife Sanctuary and Ranikhet ridges has documented high-altitude dispersal of the Bengal tiger (*Panthera tigris tigris*) up to 2,400 meters from the Corbett-Ramnagar landscape. Herbivores include the Himalayan goral (*Naemorhedus goral*) on precipitous rocky scarps (900–2,500 m), Indian muntjac or barking deer (*Muntiacus muntjak*), sambar (*Rusa unicolor*), Himalayan serow (*Capricornis sumatraensis thar*) in dense moist oak-ringal ravines of Binsar and Mornaula, and wild boar (*Sus scrofa*). Meso-carnivores and lesser mammals comprise the Asiatic black bear (*Ursus thibetanus*), yellow-throated marten (*Martes flavigula*), leopard cat (*Prionailurus bengalensis*), golden jackal (*Canis aureus*), red giant flying squirrel (*Petaurista petaurista*), rhesus macaque (*Macaca mulatta*), and Tarai gray langur (*Semnopithecus hector*).
- **Birds (Avifauna):** Designated an Important Bird and Biodiversity Area (IBA), Binsar Wildlife Sanctuary and the Chaubatia-Ranikhet oak belt support over 200 resident and migratory bird species (Upper Pindar & Ranikhet KBA). Galliformes include the Kalij pheasant (*Lophura leucomelanos*), Koklass pheasant (*Pucrasia macrolopha*) on upper Binsar ridges, Cheer pheasant (*Catreus wallichii*) on steep grassy slopes near Ranikhet, and winter-visiting Himalayan monal (*Lophophorus impejanus*) descending to Bhatrojkhan and Pandavkholi. Specialist passerines and raptors include the striated laughingthrush (*Grammatoptila striata*), red-billed blue magpie (*Urocissa erythroryncha*), maroon oriole (*Oriolus traillii*), rufous-bellied woodpecker (*Dendrocopos hyperythrus*), black eagle (*Ictinaetus malaiensis*), mountain hawk-eagle (*Nisaetus nipalensis*), and Himalayan griffon vultures (*Gyps himalayensis*).
- **Reptiles & Amphibians:** Includes the Himalayan pit viper (*Gloydius himalayanus*) in rocky oak litter (1,500–2,500 m), Indian cobra (*Naja naja*), common krait (*Bungarus caeruleus*), Oriental rat snake (*Ptyas mucosa*), Himalayan agama (*Paralaudakia himalayana*) on slate outcrops, and Bengal monitor (*Varanus bengalensis*) in lower Bhikiyasain valleys. Amphibian guilds in mountain streams (*gadheras*) feature the paa frog (*Nanorana minica*), Himalayan bullfrog (*Hoplobatrachus crassus*), and skittering frog (*Euphlyctis cyanophlyctis*).
- **Insects & Butterflies:** Oak-rhododendron canopies in Binsar and Sitlakhet support the Common Peacock (*Papilio bianor polyctor*—State Butterfly), Great Windmill (*Byasa dasarada*), Siren (*Hestina persimilis*), Blue Admiral (*Kaniska canace*), Common Map (*Cyrestis thyodamas*), giant Saturniid moon moths (*Actias selene*), and wild Himalayan cliff bees (*Apis laboriosa*).
- **Aquatic Species:** The Western Ramganga (between Chaukhutiya and Marchula) and lower Kosi support spawning runs of the endangered Golden mahseer (*Tor putitora*), snow trout (*Schizothorax richardsonii*), stone loaches (*Nemacheilus* spp.), and sucker-head hillstream catfishes (*Glyptothorax pectinopterus*), with smooth-coated otters (*Lutrogale perspicillata*) inhabiting deep pools in the Marchula gorge.

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** Moist north-facing *Quercus-Rhododendron* ravines harbor moisture-dependent Himalayan serow, Koklass pheasants, and *Nanorana* frogs, whereas dry south-facing *Pinus roxburghii* slopes with *Themeda-Arundinella* grass understories sustain Himalayan goral and Cheer pheasant. During winter (December–February), altitudinal migrants including the Himalayan monal, forktails, accentors, and wallcreepers descend from the Pindar-Sunderdhunga divide into Almora's middle-hill valleys.
- **Statutory Conservation Status:** Key Schedule I species under the Wildlife (Protection) Act, 1972 (as amended in 2022) in Almora include the Bengal tiger (IUCN Endangered; CITES Appendix I), Indian leopard (IUCN Vulnerable; CITES Appendix I), Asiatic black bear (IUCN Vulnerable; CITES Appendix I), Himalayan serow (IUCN Vulnerable; CITES Appendix I), Himalayan goral (IUCN Near Threatened; CITES Appendix I), and Cheer pheasant (IUCN Vulnerable; CITES Appendix I).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Identified as one of Uttarakhand's top three human-wildlife conflict (HWC) hotspots alongside Pauri and Tehri (Drivers of HWC in Uttarakhand, ResearchGate HWC Analysis). Conflict is concentrated in Dhauladevi, Bhaisiyachana, Hawalbagh, Lamgarha, Takula, and Syaldey blocks, where rural out-migration and abandoned terraced fields (*banjar* land) colonized by invasive *Lantana camara* and *Ageratina adenophora* provide daytime ambush cover for leopards right beside village homes. Simultaneously, rhesus macaques (*Macaca mulatta*) and wild boars (*Sus scrofa*) raid finger millet (*mandua*), maize, and fruit orchards across Someshwar and Chaubatia. Mitigation includes Quick Response Teams (QRTs), bio-fencing with lemongrass and agave, solar street lighting, and the Monkey Sterilization Centre at Almora.
- **Environmental Threats:** Summer crown and surface fires fueled by resinous Chir pine needles (*pirul*) represent the primary ecological threat, periodically burning into Binsar Wildlife Sanctuary (Forest Fire Threat Analysis). Additionally, the progressive drying of headwater springs has reduced dry-season baseflows in the non-glacial Kosi River by over 60%, threatening spawning riffles of Golden mahseer and accelerating the replacement of moisture-retaining Banj oak by fire-adapted Chir pine.

2. Bageshwar District (Kumaon Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Carved out of northern Almora in 1997, Bageshwar district (2,241 sq km; official portal bageshwar.nic.in) spans the transition from subtropical river valleys (850 m at Bageshwar town) to the southern glacial wall of the Nanda Devi massif (6,861 m at Nanda Kot and 6,663 m at Panwali Dwar). Its northern high-altitude catchments—encompassing the Pindari, Kafni, Sunderdhunga, and Maiktoli glacial valleys—form the southern buffer zone of the Nanda Devi Biosphere Reserve and contain the globally designated **Upper Pindar Catchment Key Biodiversity Area (KBA 18452)** (Upper Pindar KBA Factsheet).
- **Forest Cover & Types:** Administered by the Bageshwar Forest Division, total forest cover reaches 1,262.69 sq km (56.34% of district area), consisting of 162.39 sq km of Very Dense Forest, 761.61 sq km of Moderately Dense Forest, and 338.69 sq km of Open Forest (ISFR Vol II). Sub-tropical Chir pine (*Pinus roxburghii*) dominates the Garur, Kapkot, and Kanda valleys (900–1,800 m). Above 1,800 meters around Kausani, Vijaypur, and Loharkhet, dense Moist Temperate Broadleaf and Coniferous Forests of Banj oak (*Quercus leucotrichophora*), Moru oak (*Quercus floribunda*), Kharsu oak (*Quercus semecarpifolia*), horse chestnut (*Aesculus indica*), alder (*Alnus nepalensis*), and Himalayan silver fir (*Abies pindrow*) thrive, underlain by dense thickets of Ringal dwarf bamboo (*Drepanostachyum falcatum* and *Thamnocalamus spathiflorus*). Between 3,200 and 4,500 meters, sub-alpine Bhojpatra (*Betula utilis*) and *Rhododendron campanulatum* krummholz open into extensive alpine meadows (**Pindari Bugyal**, **Kafni Bugyal**, **Sunderdhunga**, **Baluni Bugyal**, and **Phurkia**).
- **Aquatic Ecosystems & Water Bodies:** Defined by three major river systems: (1) the **Saryu River**, originating at Sarmool on the southern slopes of Nanda Kot and flowing through Kapkot to Bageshwar; (2) the **Gomti River**, originating in the Bhatkot hills near Debra and joining the Saryu at the sacred **Bageshwar Prayag** (850 m); and (3) the glacier-fed **Pindar River**, originating from the Pindari Glacier (3,627 m) and joined by the Kafni and Sunderdhunga rivers near Khati before flowing west into Chamoli. High-altitude glacial tarns include **Devi Kund** (~4,300 m near Sunderdhunga), **Sukram Tal**, and **Bhadrapati Tal**.

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 2. Bageshwar District:



Mammal: Himalayan Tahr (Hemitragus jemlahicus) — Upper Pindar & Dwali cliffs



Mammal: Himalayan Serow (Capricornis sumatraensis thar) — Sunderdhunga Ringal ravines



Bird: Himalayan Monal (Lophophorus impejanus) — Phurkia and Pindari alpine slopes



Fish: Golden Mahseer (Tor putitora) — Saryu-Gomti confluence at Baijnath & Bageshwar

- **Mammals:** The upper glacial basins of Pindari, Kafni, and Sunderdhunga (3,200–5,200 m) support the Snow leopard (*Panthera uncia*) and Himalayan brown bear (*Ursus arctos isabellinus*), while the Indian leopard (*Panthera pardus fusca*) and Asiatic black bear (*Ursus thibetanus*) inhabit the temperate forests of Khati, Dwali, Kausani, and Kapkot. According to the Key Biodiversity Areas database, the Upper Pindar Catchment supports some of the largest herds of Himalayan tahr (*Hemitragus jemlahicus*) in India across the steep cliffs of Dwali and Phurkia (2,500–4,200 m), alongside Bharal (*Pseudois nayaur*) above 3,600 meters, Alpine musk deer (*Moschus chrysogaster*) in birch-rhododendron scrub (2,800–4,000 m), Himalayan serow (*Capricornis sumatraensis thar*) in dense Ringal bamboo gorges, Himalayan goral, sambar, barking deer, Royle's pika (*Ochotona roylei*), and yellow-throated marten.
- **Birds (Avifauna):** The Upper Pindar Catchment KBA and Kausani-Baijnath corridor harbor threatened Western Himalayan pheasants: the Western tragopan (*Tragopan melanocephalus*) and Satyr tragopan (*Tragopan satyra*) in dense Ringal-oak understories of Sunderdhunga and Pindar, the State Bird Himalayan monal (*Lophophorus impejanus*) at Phurkia and Jatoli, Cheer pheasant (*Catreus wallichii*) on steep grassy slopes near Wachham and Loharkhet, Koklass pheasant, and Himalayan snowcock (*Tetraogallus himalayensis*). Raptors include the Bearded vulture (*Gypaetus barbatus*), golden eagle (*Aquila chrysaetos*), and Himalayan griffon, alongside riverine specialists (little and spotted forktails, brown dipper, and crested kingfisher) along the Saryu and Pindar.
- **Reptiles & Amphibians:** Includes the Himalayan pit viper (*Gloydius himalayanus*) recorded up to 3,200 meters near Dwali, Himalayan keelback (*Herpetoreas platyceps*), common rat snake, Kashmir rock agama (*Laudakia tuberculata*) along the Pindar gorge, Sikimmese skink (*Asymblepharus sikimmensis*), Liebig's paa frog (*Nanorana liebigii*), and torrent frogs (*Amolops* spp.).
- **Insects & Butterflies:** Sub-alpine meadows at Kafni and Pindari support high-altitude Parnassians including the Common Red Apollo (*Parnassius epaphus*), Regal Apollo (*Parnassius charltonius*), Ladakh Tortoiseshell (*Aglais ladakensis*), Painted Lady (*Vanessa cardui*), and Clouded Yellows (*Colias fieldii*).
- **Aquatic Species:** The Saryu-Gomti confluence at Bageshwar and the Baijnath Barrage reservoir on the Gomti serve as a premier sanctuary for dense schools of Golden mahseer (*Tor putitora*) and snow trout (*Schizothorax richardsonii* and *Schizothorax plagiostomus*), where temple protection at Baijnath preserves large broodstock cohorts alongside hillstream catfishes (*Pseudecheneis sulcata*).

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** Dense Ringal bamboo (*Thamnocalamus* and *Drepanostachyum*) understories beneath *Quercus semecarpifolia* and *Abies pindrow* (2,400–3,300 m) form an obligatory micro-habitat for Himalayan serow and tragopans. During heavy winter snowfall (December–March), Himalayan tahr and Alpine musk deer migrate down-slope from Zero Point and Phurkia (3,600 m) to the sheltered cliffs and oak-ringal ravines around Dwali and Khati (2,200–2,600 m), tracked closely by snow leopards.
- **Statutory Conservation Status:** Priority protected taxa include the Alpine musk deer (IUCN Endangered; CITES Appendix I; WPA Schedule I), Snow leopard (IUCN Vulnerable; CITES Appendix I; WPA Schedule I), Western tragopan

(IUCN Vulnerable; CITES Appendix I; WPA Schedule I), Himalayan tahr (IUCN Near Threatened; WPA Schedule I), and Golden mahseer (IUCN Endangered).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Human-leopard conflict is concentrated in Garur, Bageshwar, Kanda, and lower Kapkot blocks, where leopards ambush livestock and villagers collecting fodder. In high-elevation villages of Khati, Wachham, Jatoli, and Jhuni (2,200–2,700 m), Asiatic black bears (*Ursus thibetanus*) raid autumn maize, amaranth, and buckwheat crops and attack migratory sheep-goat flocks. Mitigation includes solar-powered wire fencing around remote hamlets, predator-deterrent lights, and prompt ex-gratia compensation.
- **Environmental Threats:** Unregulated commercial harvesting of caterpillar fungus (*Ophiocordyceps sinensis*, locally *Keeda Jadi*) across the Pindari, Kafni, and Sunderdhunga bugyals during May–June—the critical fawning and nesting season for musk deer, tahr, and monal—causes severe trampling of alpine turf, cutting of slow-growing *Betula utilis* for fuelwood, and opportunistic poaching (Himalayan Meadows Governance). Simultaneously, open-cast **soapstone (talc) mining** in Kanda and Kapkot tehsils triggers land subsidence and heavy siltation into the Saryu watershed, while rapid recession of the Pindari Glacier alters downstream hydrology.

3. Chamoli District (Garhwal Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** As Uttarakhand's largest district by area (8,030 sq km), Chamoli district anchors the Greater and Trans-Himalayan Garhwal landscape and encompasses two UNESCO World Heritage National Parks and India's largest Western Himalayan wildlife sanctuary (UNESCO Nanda Devi & Valley of Flowers):
  - **Nanda Devi National Park (624.60 sq km):** Established in 1982 and inscribed as a UNESCO World Heritage Site in 1988, protecting the glacier-ringed Nanda Devi Sanctuary (rimmed by 12 peaks >6,400 m, culminating in Nanda Devi West at 7,816 m) drained via the Rishi Ganga gorge. Core entry is strictly prohibited to preserve its undisturbed baseline ecology.
  - **Valley of Flowers National Park (87.50 sq km):** Notified in 1982 and added to the UNESCO World Heritage site in 2005 (Uttarakhand Tourism Valley of Flowers), occupying the Pushpawati glacial valley (3,350–6,719 m) in the transition zone between the Zanskar and Great Himalayan ranges. Together with Nanda Devi NP, these two core zones sit within the **Nanda Devi Biosphere Reserve** (6,407.03 sq km total, including a 5,148.57 sq km buffer and 546.34 sq km transition zone).
  - **Kedarnath Wildlife Sanctuary (975.20 sq km total):** Notified on 21 January 1972 (UKPSC Protected Area Analysis), its eastern expanse covers Chamoli's Gopeshwar, Mandal, Chopta-Baniyakund, and Urgam valleys and houses the **Kanchula Korak Musk Deer Breeding Centre**.
- **Forest Cover & Types:** Managed by Badrinath Forest Division, Kedarnath Wildlife Sanctuary FD (Gopeshwar), and Nanda Devi National Park FD (Joshimath), Chamoli holds 2,710.11 sq km of forest cover (33.75% of district area, as 66.25% lies above the 3,600 m timberline), including 443.00 sq km of Very Dense Forest, 1,580.68 sq km of Moderately Dense Forest, and 686.43 sq km of Open Forest (ISFR Vol II). Sub-tropical Chir pine lines the lower Alaknanda and Pindar gorges (800–1,700 m); moist temperate broadleaf forests of Banj, Moru, and Kharsu oak, maple, walnut, yew (*Taxus wallichiana*), and Deodar cloak Mandal, Urgam, and Joshimath slopes (1,800–3,000 m); sub-alpine birch (*Betula utilis*) and blue pine (*Pinus wallichiana*) fringe the treeline (3,000–3,600 m); and iconic alpine **Bugyals**—**Bedni Bugyal** (3,354 m), **Ali Bugyal**, **Auli-Gorson Bugyal**, **Bagji Bugyal**, **Rudranath-Panar Bugyal**, and **Kuari Pass**—blanket the 3,200–4,500 m zone, while cold-desert Trans-Himalayan steppe scrub (*Caragana-Lonicera-Eurotia*) occurs in the **Niti and Mana valleys**.
- **Aquatic Ecosystems & Water Bodies:** Cradle of the **Alaknanda River** (rising from the Satopanth and Bhagirath Kharak glaciers near Mana at 3,800 m), which forms three of the sacred Panch Prayags within Chamoli: **Vishnuprayag** (with the Dhauliganga/Rishi Ganga at 1,372 m), **Nandaprayag** (with the Nandakini River at 870 m), and **Karnaprayag** (with the Pindar River at 788 m), alongside the Birahi Ganga and Balkhila River. High-altitude lakes include **Hemkund Lokpal Lake** (4,329 m), **Roopkund ("Mystery Lake" at 5,029 m)**, **Satopanth Tal** (4,600 m), **Homkund**, **Kagbhusandi Tal**, **Benital**, and **Gohna Tal**.

2. Wildlife & Faunal Catalogue

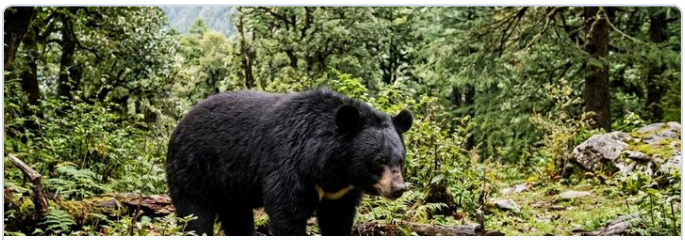
Featured Wildlife & Faunal Species Photograph Gallery — 3. Chamoli District:



Mammal: Snow Leopard (Panthera uncia) — Nanda Devi National Park & Niti Valley



Mammal: Alpine Musk Deer (Moschus chrysogaster) — Valley of Flowers & Kedarnath WLS



Mammal: Himalayan Black Bear (Ursus thibetanus laniger) — Mandal-Chopta temperate forests



Bird: Himalayan Monal (Lophophorus impejanus) — Tungnath & Valley of Flowers Bugyals

- **Mammals:** Nanda Devi National Park, the upper Dhauliganga (Niti-Malari), and Mana valleys support dense guilds of Snow leopard (*Panthera uncia*), Himalayan brown bear (*Ursus arctos isabellinus*), and Tibetan wolf (*Canis lupus chanco*). Alpine and Trans-Himalayan ungulates include large herds of Bharal (*Pseudois nayaur*, 3,400–5,200 m), Himalayan tahr (*Hemitragus jemlahicus*, 2,500–4,200 m at Lata Kharak), Alpine musk deer (*Moschus chrysogaster*, 2,800–4,200 m), and occasional Kiang (*Equus kiang*) and Tibetan argali (*Ovis ammon hodgsoni*) in the northern Niti-Rimkhim borderlands. Temperate forests of Mandal-Chopta and Joshimath support the Indian leopard, Asiatic black bear, Himalayan serow, Himalayan goral, sambar, barking deer, yellow-throated marten, red fox (*Vulpes vulpes montana*), woolly flying squirrel (*Eupetaurus cinereus*), and Royle's pika.
- **Birds (Avifauna):** The **Mandal-Chopta valley** (1,500–3,600 m) inside Kedarnath Wildlife Sanctuary is a premier Himalayan birding hotspot supporting dense breeding populations of the State Bird Himalayan monal (*Lophophorus impejanus*), Satyr tragopan (*Tragopan satyra*), Koklass pheasant (*Pucrasia macrolopha*), Kalij pheasant, and snow partridge (*Lerwa lerwa*). High-altitude species across Nanda Devi and Valley of Flowers include the Bearded vulture, Himalayan griffon, golden eagle, Himalayan snowcock, grandala (*Grandala coelicolor*), red-fronted rosefinch, fire-tailed myzornis (*Myzornis pyrrhoura*), golden bush robin, wallcreeper (*Tichodroma muraria*), and ibisbill (*Ibidorhyncha struthersii*).
- **Reptiles & Amphibians:** Includes the high-altitude Himalayan pit viper (*Gloydius himalayanus*) recorded up to 3,600 meters in the Valley of Flowers and Lata Kharak, Himalayan agama (*Paralaudakia himalayana*) in the dry Dhauliganga gorge (Malari-Niti), Himalayan toad (*Duttaphrynus himalayanus*), and Liebig's paa frog (*Nanorana liebigii*) in Mandal streams.
- **Insects & Butterflies:** Valley of Flowers National Park hosts an extraordinary pollinator guild across 600+ alpine angiosperm species (*Brahmakamal Saussurea obvallata*, *Meconopsis aculeata* blue poppy, *Primula*, *Cypripedium* orchids), featuring Common Yellow Swallowtail (*Papilio machaon*), Common Blue Apollo (*Parnassius hardwickii*), Common Red Apollo (*Parnassius epaphus*), Himalayan Jester (*Symbrenthia hypselis*), Queen of Spain Fritillary, and endemic bumblebees (*Bombus haemorrhoidalis*).
- **Aquatic Species:** The Alaknanda, Pindar, Nandakini, and Balkhila rivers support snow trout (*Schizothorax richardsonii*, *Schizothoraichthys progastus*), Golden mahseer (*Tor putitora*, migrating upstream to Karnaprayag and Nandaprayag during monsoon), hillstream loaches (*Schistura* spp.), and sucker catfishes (*Glyptothorax* and *Pseudecheneis*).

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** In autumn (October–November), Bharal and Snow leopard descend from the 4,800-meter moraines of Nanda Devi and Satopanth into wind-swept rain-shadow cliffs of Malari, Lata, and Niti (2,800–3,500 m). Simultaneously, Alpine musk deer occupy sub-alpine *Betula-Rhododendron* krummholz (3,000–3,800 m), while Himalayan monal flocks descend from Tungnath and Valley of Flowers meadows into the snow-free oak-maple forests of Mandal and Baniyakund (1,800–2,400 m).
- **Statutory Conservation Status:** Protected Schedule I species include the Snow leopard (IUCN Vulnerable; CITES Appendix I), Alpine musk deer (IUCN Endangered; CITES Appendix I), Himalayan brown bear (IUCN Critically

Endangered in India; CITES Appendix I), Woolly flying squirrel (IUCN Endangered), Himalayan serow (IUCN Vulnerable; CITES Appendix I), and Himalayan monal (CITES Appendix I).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Driven by Asiatic black bears (*Ursus thibetanus*) and Indian leopards. Across Joshimath, Dasholi, Pokhari, Ghat, Dewal, and Tharali blocks, black bears raid maize, apple, and amaranth fields and break into winter cattle sheds (*gaushalas*), causing severe human maulings during October–November hyperphagia before winter denning. Leopards prey on livestock and dogs along the Gopeshwar, Karnaprayag, and Gauchar valley settlements, while snow leopards and Tibetan wolves occasionally depredate corralled sheep and goats in Niti-Malari.
- **Environmental Threats:** Cascades of run-of-the-river hydroelectric projects—including Vishnuprayag (400 MW), Tapovan-Vishnugad (520 MW, struck by the February 2021 Rishi Ganga rock-ice avalanche), and Vishnugad-Pipalkoti (444 MW)—divert lean-season river discharge into headrace tunnels, desiccating riverbeds critical for *Schizothorax* and *Tor putitora* (Char Dham & Hydroelectric Crisis). Slope cutting for the Char Dham All-Weather Road (NH-7), land subsidence in Joshimath, and helicopter noise (>80 dB) over Badrinath and Hemkund Sahib impact sub-alpine fauna. Bio-geotextile coir matting is actively deployed on Bedni and Ali Bugyals to arrest alpine turf erosion (Alpine Meadows Conservation).

4. Champawat District (Kumaon Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Situated on the southeastern frontier of Kumaon along the Indo-Nepal border, Champawat district (1,766 sq km; official portal champawat.nic.in) is Uttarakhand's smallest district by area yet ranks second in the state in proportional forest cover (69.32% in ISFR 2021; 69.06% in ISFR 2023) (ISFR District Ranking). Its southwestern Bhabar-Shivalik belt (Tanakpur-Chalthi-Danda sector) encompasses the eastern core and buffer ranges of **Nandhaur Wildlife Sanctuary** (269.96 sq km total, notified in 2012; official portal nandhaurwildlife.uk.gov.in). Nandhaur forms a crucial link in the **Shivalik Elephant Reserve** and the **Boom-Brahmadev transboundary wildlife corridor**, connecting the Ramnagar-Haldwani-Terai East forests of India with **Shuklaphanta National Park** in Nepal across the Sharda (Mahakali) River (Nandhaur Corridor Study).
- **Forest Cover & Types:** Administered by the Champawat Forest Division and eastern ranges of Haldwani Forest Division (Boom, Sharda, and Danda ranges), the district supports 1,224.00 sq km of forest cover (339.09 sq km Very Dense, 578.96 sq km Moderately Dense, and 305.95 sq km Open Forest) (ISFR 2023 Vol II). From 200 to 1,000 meters in the Tanakpur-Banbasa Bhabar and Shivalik hills, Northern Tropical Moist Deciduous Sal Forests (*Shorea robusta*, *Terminalia alata*, *Haldina cordifolia*) and riverine Khair-Sissoo (*Senegalia catechu–Dalbergia sissoo*) dominate. Between 1,000 and 1,700 meters around Chalthi and Sukhidhang, sub-tropical Chir pine occurs. From 1,700 to 2,450 meters across Champawat, Lohaghat, Abbott Mount (2,100 m), Mayawati Ashram, and Devidhura, lush Moist Temperate Banj and Rianj Oak (*Quercus leucotrichophora*, *Quercus lanata*), Burans, and sacred Deodar (*Cedrus deodara*) groves flourish.
- **Aquatic Ecosystems & Water Bodies:** Bounded on the east by the **Sharda (Kali/Mahakali) River**, which carves the deep **Pancheshwar and Poornagiri gorges** before debouching onto the plains at Boom and Banbasa Barrage. Interior rivers include the **Lohawati River** (draining Lohaghat into the Kali), **Ladhiya River** (joining the Kali at Chuka), **Saryu River** (forming the northern border with Pithoragarh before joining the Kali at **Pancheshwar Prayag**), and the upper headwaters of the **Nandhaur River** (near Durgapipal). Key lentic wetlands include **Shyamala Tal** (1,300 m near Sukhidhang) and **Jhilmil Tal** (near Tanakpur-Purnagiri).

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 4. Champawat District:



Mammal: Asian Elephant (Elephas maximus) — Boom-Nandhaur transboundary corridor



Bird: Great Hornbill (Buceros bicornis) — Sub-Himalayan Sal forests of Nandhaur WLS



Reptile: King Cobra (Ophiophagus hannah) — Ladhiya and Chalthi bamboo gorges



Fish: Golden Mahseer (Tor putitora) — Pancheshwar confluence & Kali River

- **Mammals:** Exhibits direct overlap between lowland Terai megafauna and montane Himalayan species. The Boom, Sharda, and Nandhaur ranges support resident and transboundary Bengal tiger (*Panthera tigris tigris*) and Indian elephant (*Elephas maximus indicus*) herds, with camera traps documenting tigers ascending into temperate oak forests at 2,000+ meters near Lohaghat and Devidhura—echoing the historical ecology of the Champawat Tiger shot by Jim Corbett in 1907. The Indian leopard, sloth bear (*Melursus ursinus*, in lower Tanakpur-Boom Bhabar), and Asiatic black bear (*Ursus thibetanus*, in Lohaghat-Pati oak belts) coexist with sambar, chital (*Axis axis*), barking deer, Himalayan goral, Himalayan serow, wild boar, rusty-spotted cat (*Prionailurus rubiginosus*), yellow-throated marten, and smooth-coated otter (*Lutrogale perspicillata*) along the Kali and Ladhiya rivers.
- **Birds (Avifauna):** Abbott Mount, Mayawati, Shyamala Tal, and the Boom-Nandhaur corridor host over 300 avian species. Sub-Himalayan Bhabar forests harbor the Great hornbill (*Buceros bicornis*), Oriental pied hornbill, red-headed trogon, dollarbird, and collared falconet, while temperate ridges around Abbott Mount and Devidhura support Kalij pheasant, Cheer pheasant, white-crested laughingthrush, blue-winged minla, and steppes and cinereous vultures (*Aegypius monachus*).
- **Reptiles & Amphibians:** Moist Bhabar-Shivalik ravines of Chalthi, Ladhiya, and Nandhaur are a stronghold of the King cobra (*Ophiophagus hannah*), nesting in bamboo and Sal leaf-litter mounds up to 1,600 meters. Other herpetofauna includes the Burmese python (*Python bivittatus*), Russell's viper, banded krait (*Bungarus fasciatus*), Mugger crocodile (*Crocodylus palustris*) in the Sharda Barrage and lower Ladhiya pools (Terai Marsh Crocodile Study), yellow monitor (*Varanus flavescens*), and marbled balloon frog (*Uperodon systoma*).
- **Insects & Butterflies:** Shyamala Tal and Chalthi riverine corridors support Paris Peacock (*Papilio paris*), Red Helen (*Papilio helenus*), Five-bar Swordtail (*Graphium antiphates*), Tawny Rajah (*Charaxes bernardus*), Popinjay (*Stibochiona nicea*), and Orange Oakleaf (*Kallima inachus*).
- **Aquatic Species:** The confluence of the Saryu and Kali rivers at **Pancheshwar** and the **Chuka-Boom** stretch of the Sharda River represent the world's premier trophy habitat for the Golden mahseer (*Tor putitora*), where specimens exceeding 25–40 kg ascend from the Sharda Barrage to spawn in gravel-cobble runs, alongside Giant devil catfish (*Bagarius yarrelli*, locally *Goonch*), snow trout (*Schizothorax richardsonii*), and *Tor tor*.

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** Champawat's steep north-south gradient from the 220-meter Banbasa Terai to the 2,200-meter Abbott Mount crest within 35 km enables rapid vertical movement of large carnivores. Elephants utilize the Boom-Brahmadev riparian corridor along the Sharda River to cross seasonally between Nandhaur Wildlife Sanctuary and Nepal's Shuklaphanta National Park, while Golden mahseer require unobstructed longitudinal river connectivity from the Sharda plains into the Saryu and Ladhiya spawning tributaries.
- **Statutory Conservation Status:** Priority protected species include the Bengal tiger (IUCN Endangered; CITES Appendix I; WPA Schedule I), Indian elephant (IUCN Endangered; CITES Appendix I; WPA Schedule I), King cobra (IUCN Vulnerable; CITES Appendix II; WPA Schedule I), Great hornbill (IUCN Vulnerable; CITES Appendix I; WPA Schedule I), and Golden mahseer (IUCN Endangered).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** In the southern plains of Tanakpur and Banbasa (adjacent to Boom Range), Indian elephant herds raid paddy, wheat, and sugarcane fields, while in the hill blocks of Champawat, Lohaghat, Barakot, and Pati, Indian leopard attacks on livestock and humans are recurrent. Occasional tiger livestock kills occur around Chalthi, Sukhidhang, and Danda villages bordering Nandhaur WLS. Mitigation includes solar elephant-proof fencing in Banbasa, camera-trap monitoring of dispersing tigers, and community eco-tourism guide training (Nandhaur Eco-Tourism).
- **Environmental Threats:** The proposed **Pancheshwar Multipurpose Dam Project** (315-meter rockfill dam on the Mahakali/Sharda River at the Saryu confluence) threatens to submerge the free-flowing spawning grounds of the Golden mahseer and *Bagarius yarrelli* as well as >120 sq km of riverine forest. Additionally, intensive **River Bed**

Material (RBM) boulder mining in the Sharda and Nandhaur rivers degrades riparian corridors, and widening of the **Tanakpur-Pithoragarh All-Weather Highway (NH-9)** has caused chronic slope destabilization at Swala and Chalthi.

5. Dehradun District (Garhwal Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Dehradun district (3,088 sq km) occupies the tectonic Doon Valley bounded by the Shivalik Range to the south, the Lesser Himalayan Mussoorie-Jaunsar-Bawar ranges to the north, the Ganges to the east, and the Yamuna-Tons to the west. It encompasses three statutory protected areas and serves as India's premier hub of wildlife science (housing the Wildlife Institute of India, Forest Survey of India, Forest Research Institute, and Zoological Survey of India Northern Regional Centre):
  - **Rajaji National Park & Tiger Reserve (Western & Central Sector):** Amalgamated in 1983 and notified as a Tiger Reserve in 2015 (820.42 sq km core, 1,075.17 sq km total), its **Ramgarh, Kansrau, Motichur, Phandowala, and western Gohri ranges** lie in Dehradun district along the northern Shivalik dip-slopes and Doon floor (300–1,000 m).
  - **Mussoorie Wildlife Sanctuary (Benog Wildlife Sanctuary, 10.82 sq km):** Notified on 2 September 1993, situated 11 km west of Mussoorie in the Aglar River catchment (1,400–2,250 m), protecting temperate oak-pine ridges and historic habitat of the critically endangered Himalayan quail (*Ophrysia superciliosa*).
  - **Asan Barrage (Asan Conservation Reserve, 4.44 sq km):** Declared India's first wetland Conservation Reserve in 2005 and designated **Uttarakhand's first Ramsar Site** in October 2020 (Ramsar Sites in India), situated at the confluence of the Asan and Yamuna rivers at Dhalipur (389 m).
- **Forest Cover & Types:** Managed by Dehradun FD, Mussoorie FD, Chakrata FD, Kalsi Soil Conservation FD, and Rajaji Tiger Reserve Division, the district maintains 1,611.58 sq km of forest cover (52.19% of district area)—holding 663.25 sq km of Very Dense Forest (second highest in the state after Nainital), 636.33 sq km of Moderately Dense Forest, and 312.00 sq km of Open Forest (ISFR District Data). In the Doon Valley (Thano, Lachhiwala, Barkot, Timli, and Kansrau), Moist Siwalik and Doon Valley Sal Forests (*Shorea robusta*, Type 3C/C2a) and relict tropical freshwater swamp forests (**Golatappar and Karwapani *Calamus* cane and *Diospyros peregrina* swamps** in Barkot/Kansrau) prevail. In the Mussoorie hills (1,500–2,270 m), Banj oak, Burans, and Deodar occur, while the northern **Chakrata-Deoban-Kanasar-Moila Top (Jaunsar-Bawar)** belt (2,000–3,080 m) supports old-growth Himalayan Deodar (*Cedrus deodara*), Moru and Kharsu oak, West Himalayan spruce (*Picea smithiana*), and silver fir (*Abies pindrow*) surrounding **Moila Bugyal (Budher)**.
- **Aquatic Ecosystems & Water Bodies:** Hydrologically divided by the central Dehradun ridge into two basins: the **Eastern Doon**, drained by the **Song River** and its spring-fed tributaries—the **Suswa River**, **Rispana**, **Bindal**, **Jakhan**, and **Baldi (Sahastradhara)**—joining the **Ganges River** near Raiwala/Satyanarayan; and the **Western Doon**, drained by the **Asan River** and **Sitla Rao**, flowing west into the **Yamuna River** at Asan Barrage. In northern Jaunsar-Bawar, the **Tons River** merges with the Yamuna at **Kalsi-Dakpathar**, joined by the **Aglar River** upstream near Yamuna Bridge.

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 5. Dehradun District:



Mammal: Asian Elephant (Elephas maximus) — Rajaji National Park Western Ranges



Mammal: Indian Leopard (Panthera pardus fusca) — Mussoorie Wildlife Sanctuary



Bird: Great Hornbill (Buceros bicornis) — Thano and Golatappar swamp forests



Reptile: King Cobra (Ophiophagus hannah) — Kansrau and Lachhiwala Sal belts

- **Mammals:** Rajaji National Park marks the northwesternmost global distribution limit of both the Bengal tiger (*Panthera tigris tigris*) and the Indian elephant (*Elephas maximus indicus*) (PMF IAS Protected Areas). Following the **WII-NTCA Tiger Translocation Project (2020–2024)** into Western Rajaji (Motichur-Kansrau), resident breeding tigers occupy both sides of the Ganges alongside >350 elephants, dense populations of Indian leopard, sloth bear (*Melursus ursinus*), striped hyena (*Hyaena hyaena*) in Shivalik *raos*, sambar, chital, nilgai (*Boselaphus tragocamelus*), barking deer, Himalayan goral, Indian pangolin (*Manis crassicaudata*), and smooth-coated otter. In northern **Chakrata-Deoban** (2,200–3,000 m), Asiatic black bear, Himalayan serow, Himalayan tahr (upper Tons cliffs), leopard cat, yellow-throated marten, and red giant flying squirrel thrive.
- **Birds (Avifauna):** Over 500 recorded bird species across three distinct zones: (1) **Asan Barrage (Ramsar Site):** 330+ species including 90+ winter Palearctic waterbird migrants—ruddy shelduck (*Tadorna ferruginea*, >4,000 individuals), red-crested pochard, ferruginous duck, critically endangered Baer's pochard (*Aythya baeri*), red-headed vulture (*Sarcogyps calvus*), white-rumped vulture (*Gyps bengalensis*), Pallas's fish eagle, and osprey; (2) **Rajaji-Thano-Golatappar Swamp:** Great hornbill (*Buceros bicornis*), white-naped woodpecker (*Chrysocolaptes festivus*), Sultan tit, and silver-eared mesia; and (3) **Mussoorie Wildlife Sanctuary (Benog) & Deoban (Chakrata):** Cheer pheasant, Koklass pheasant, Kalij pheasant, Himalayan monal (Deoban-Mundali), white-throated tit, and Bearded vulture.
- **Reptiles & Amphibians:** Supports the King cobra (*Ophiophagus hannah*), Burmese python (*Python bivittatus*, in Kansrau, Lachhiwala, and FRI estate), Russell's viper, bamboo pit viper (*Craspedocephalus gramineus*), Himalayan pit viper (Chakrata), common krait, Bengal and yellow monitors, Indian flapshell turtle (*Lissemys punctata*), and crowned river turtle (*Hardella thurjii*) in the Asan-Yamuna basin. Amphibians include the Chakrata torrent frog (*Amolops chakrataensis*, described from Tiger Falls, Chakrata), fungoid frog, and tree frogs (*Polypedates maculatus*).
- **Insects & Butterflies:** The Forest Research Institute (FRI) campus, Lachhiwala Nature Park, and Deoban host over 320 documented butterfly species, including the Common Peacock (*Papilio bianor*), Siren (*Hestina persimilis*), Great Eggfly, Commander (*Moduza procris*), Common Baron, and *Zephyrus* hairstreaks in Chakrata oak woods.
- **Aquatic Species:** The Song, Suswa, Asan, Yamuna (Dakpathar-Kalsi), and Tons rivers support Golden mahseer (*Tor putitora*), Copper mahseer (*Neolissochilus hexagonolepis*), snow trout (*Schizothorax richardsonii*), *Barilius bendelisis*, *Garra gotyla*, and subterranean cave-adapted loaches in karstic springs around Sahastradhara and Budher caves.

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** East-west seasonal movement of Indian elephant and Bengal tiger between the Yamuna and Ganges depends on two pinch-point corridors: the **Chilla-Motichur Corridor** across the Ganges and the **Kansrau-Barkot-Rishikesh Corridor** across the Song River (Elephant Corridors of India 2023). Meanwhile, the **Golatappar Freshwater Swamp** in Barkot Range (360 m) preserves a relict tropical swamp micro-habitat critical for cane-dwelling pythons, hornbills, and smooth-coated otters.
- **Statutory Conservation Status:** Priority Schedule I species include the Bengal tiger (IUCN Endangered; CITES Appendix I), Indian elephant (IUCN Endangered; CITES Appendix I), Indian pangolin (IUCN Endangered; CITES Appendix I), Red-headed vulture (IUCN Critically Endangered; CITES Appendix II), Baer's pochard (IUCN Critically Endangered), and Cheer pheasant (IUCN Vulnerable; CITES Appendix I).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Peri-urban expansion of Dehradun, Rishikesh, Doiwala, and Herbertpur into forest fringes drives intense human-elephant and human-leopard conflict. In Thano, Raiwala, Harrawala, Doiwala, and Lachhiwala, elephants raid wheat, sugarcane, and paddy crops, occasionally causing human fatalities along forest roads. In the Rajpur, Galjwari, Raipur, and FRI-Cantonment interfaces, leopards prey on domestic dogs and cattle and enter residential colonies. Mitigation includes elevated wildlife underpasses on NH-7 (at Motichur-Kansrau), solar power fencing, elephant-proof trenches (EPTs), and the City Rapid Rescue Team at Malsi.
- **Environmental Threats:** Linear infrastructure projects bisect the Doon Valley: the **Delhi-Dehradun Expressway (NH-72A)** through the Ganeshpur-Mohand Shivalik ridge required a 12-km elevated wildlife corridor, while the **Haridwar-Dehradun railway track** passing through Kansrau-Motichur inside Rajaji TR has historically caused elephant train-hit mortalities (now mitigated via thermal camera warning systems and speed limits). Untreated urban sewage in the Rispana, Bindal, and Suswa rivers and invasive *Lantana camara* and *Parthenium hysterophorus* degrade downstream habitats in Rajaji Tiger Reserve.

6. Haridwar District (Garhwal Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Covering 2,360 sq km across the southern Shivalik hills, Bhabar piedmont, and upper Gangetic alluvial plains, Haridwar district (official portal haridwar.nic.in) contains two globally important protected areas and a Gangetic bird sanctuary:
  - **Rajaji National Park & Tiger Reserve (Southern & Eastern Sector):** Encompasses the **Haridwar, Ranipur, Beribara, Dholkhand, and Chillawali ranges** on the western bank of the Ganges, and the southern portion of the **Chilla Range** on the eastern bank. The **Chilla-Motichur Wildlife Corridor** at Raiwala-Bhimgoda (3.5 km across the Ganges islands) is the most critical megafaunal bottleneck in northwestern India.
  - **Jhilmil Jheel Conservation Reserve (37.84 sq km):** Notified on 14 August 2005 in the Chidiyapur Range of Haridwar Forest Division on the left bank of the Ganges, protecting a crescent-shaped saucer oxbow marsh—the last remnant of primordial Gangetic Terai tall wet grassland in western Uttarakhand (Jhilmil Jheel Avian Structure).
  - **Neel Dhara Pakshi Vihar (Bhimgoda Barrage Bird Sanctuary):** A 3.5-sq-km riverine wetland reservoir on the main channel of the Ganges at Bhimgoda Barrage in Haridwar city (Haridwar District Wildlife).
- **Forest Cover & Types:** Managed by Haridwar Forest Division (Rasiyabad, Chidiyapur, Shyampur, Khanpur, Laksar, and Roorkee ranges) and Rajaji Tiger Reserve Division, the district holds 588.00 sq km of forest cover (24.74% of geographical area: 99.00 sq km Very Dense, 315.00 sq km Moderately Dense, and 174.00 sq km Open Forest) (ISFR District Data). Along the sandstone-conglomerate ridges of the Shivaliks (Ranipur, Dholkhand, Mohand; 350–950 m), Dry Siwalik Sal Forests (Type 5B/C1a), Northern Dry Mixed Deciduous Forests (*Anogeissus latifolia*, *Lannea coromandelica*, *Terminalia bellirica*, *Boswellia serrata*), and Chir pine on narrow hogback crests occur. Across the Chidiyapur-Shyampur Gangetic floodplains and Jhilmil Jheel Conservation Reserve (200–250 m), Khair-Sissoo primary riverine successions (*Senegalia catechu*, *Dalbergia sissoo*), Semal-Rohini alluvial savannahs (*Bombax ceiba*), and tall Terai wet grasslands (*Phragmites karka*, *Saccharum spontaneum*, *Themeda arundinacea*, *Vetiveria zizanioides*) thrive.
- **Aquatic Ecosystems & Water Bodies:** The **Ganges River** exits the Shivalik gorge at Haridwar (290 m), bifurcating into the **Upper Ganga Canal** (at Bhimgoda Barrage, flowing through Roorkee) and the braided **Neel Dhara / Banganga** main stem flowing south through Shyampur, Chidiyapur, and Laksar. Ephemeral Shivalik boulder torrents (*raos*)—including **Ranipur Rao**, **Pathri Rao**, **Dholkhand Rao**, **Chillawali Rao**, and the perennial **Rawasan River**—recharge **Jhilmil Jheel**, **Baan Ganga Wetland** (Laksar), and **Solani River** marshes near Roorkee (WII Baan Ganga Report).

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 6. Haridwar District:



Mammal: Northern Swamp Deer / Barasingha (Rucervus duvaucelii) — Jhilmil Jheel Reserve



Mammal: Asian Elephant (Elephas maximus) — Rajaji Tiger Reserve Chilla & Ranipur



Reptile: Gharial (Gavialis gangeticus) — Ganges sandbars and islands near Bhimgoda



Mammal: Bengal Tiger (Panthera tigris tigris) — Chilla and Shyampur forest ranges

- **Mammals:** Jhilmil Jheel Conservation Reserve and the adjoining Baan Ganga marshes support Uttarakhand's flagship population of the vulnerable **Northern Swamp Deer or Barasingha** (*Rucervus duvaucelii duvaucelii*, 150–250 individuals) alongside hog deer (*Axis porcinus*), sambar, chital, nilgai, and barking deer. Apex predators across the Chilla, Shyampur, Chidiyapur, and Dholkhand ranges include the Bengal tiger (*Panthera tigris tigris*, with Chilla-Shyampur supporting ~45–50 tigers and dispersing individuals in Jhilmil Jheel), high-density Indian leopard, striped

hyena (*Hyaena hyaena*), sloth bear (*Melursus ursinus*), and golden jackal. Over 400 Indian elephant individuals utilize the Chilla, Ranipur, Dholkhand, and Shyampur ranges. Lesser mammals include the Indian pangolin, honey badger (*Mellivora capensis*), smooth-coated otter (*Lutrogale perspicillata*), and rare monsoon records of the Gangetic river dolphin (*Platanista gangetica*) near the Balawali-Bijnor border.

- **Birds (Avifauna):** Ornithological surveys in Jhilmil Jheel Conservation Reserve, Neel Dhara Pakshi Vihar, and Chilla have cataloged over 315 bird species. Flagship wetland and grassland species include the Swamp francolin (*Ortygornis gularis*), Finn's weaver (*Ploceus megarhynchus*, endangered Terai specialist), bristled grassbird (*Chaetornis striata*), black-necked stork (*Ephippiorhynchus asiaticus*), woolly-necked stork, lesser adjutant (*Leptoptilos javanicus*), sarus crane (*Antigone antigone*), bar-headed goose (*Anser indicus*), Pallas's fish eagle, red-headed and cinereous vultures, and Great and Oriental pied hornbills.
- **Reptiles & Amphibians:** The Ganges islands from Bhimgoda to Balawali and Jhilmil Jheel support breeding populations of the Mugger crocodile (*Crocodylus palustris*) and dispersing Gharial (*Gavialis gangeticus*). Terrestrial and semi-aquatic reptiles include the King cobra, Burmese python (*Python bivittatus*, reaching >5 m in Jhilmil Jheel reedbeds), Russell's viper, spectacled cobra, Indian softshell turtle (*Nilssonia gangetica*), three-striped roofed turtle (*Batagur dhongoka*), and yellow monitor (*Varanus flavescens*).
- **Insects & Butterflies:** Over 130 butterfly species recorded in Jhilmil Jheel and Ranipur, including the Common Mime (*Papilio clytia*), Crimson Rose (*Pachliopta hector*), Spot Swordtail (*Graphium nomius*), Baronet (*Symphaedra nais*), Danaid Eggfly (*Hypolimnas misippus*), and diverse odonates across the oxbow marshes.
- **Aquatic Species:** The Ganges at Haridwar (Har-ki-Pauri to Chilla and Bhimgoda Barrage) is a premier sanctuary for Golden mahseer (*Tor putitora*), *Tor tor*, rohu (*Labeo rohita*), catla (*Labeo catla*), kalbasu (*Labeo calbasu*), featherbacks (*Chitala chitala*), and giant catfish (*Wallago attu* and *Bagarius bagarius*).

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** The wet tall grassland (*Phragmites-Saccharum*) of Jhilmil Jheel Conservation Reserve provides the primary fawning and wallowing micro-habitat in western Uttarakhand for the **Swamp deer / Barasingha**, while the **Chilla-Motichur and Shyampur-Rawasan Corridors** enable seasonal movement of Indian elephant and Bengal tiger across the Ganges between Eastern Rajaji (Chilla-Lansdowne-Corbett) and Western Rajaji (Ranipur-Dholkhand-Motichur).
- **Statutory Conservation Status:** Key Schedule I species include Swamp deer / Barasingha (IUCN Vulnerable; CITES Appendix I), Hog deer (IUCN Endangered; CITES Appendix I), Bengal tiger (IUCN Endangered; CITES Appendix I), Indian elephant (IUCN Endangered; CITES Appendix I), Gharial (IUCN Critically Endangered; CITES Appendix I), Mugger crocodile (IUCN Vulnerable; CITES Appendix I), and Finn's weaver (IUCN Endangered).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Intense year-round human-elephant and human-leopard conflict occurs along the hard urban-industrial-agricultural boundary of BHEL Ranipur, SIDCUL Industrial Estate, Jagjeetpur, Missarpur, Pathri, Laksar, and Shyampur. Elephants emerging from Ranipur, Haridwar, and Shyampur ranges frequently enter BHEL townships, suburban colonies of Jagjeetpur, and sugarcane belts of Laksar and Khanpur. Simultaneously, leopards inhabit Pathri and Rasiyabad plantation blocks, while nilgai and wild boars raid crops in Roorkee and Laksar. Mitigation includes the Chidiyapur Wildlife Rescue and Rehabilitation Centre, solar power fencing, radio-collaring of problem tuskers, and elephant-watcher squads.
- **Environmental Threats:** The **Chilla-Motichur wildlife corridor** is severely constricted by four parallel linear barriers: the Chilla Hydropower Power Channel, NH-34/NH-7 highway traffic, the Haridwar-Dehradun electrified railway line, and ribbon urbanization along Raiwala-Motichur (Rajaji Corridor Analysis). Industrial expansion at **SIDCUL right up to the southern boundary of Ranipur Range**, mass religious gatherings (**Kumbh Mela and Kanwar Yatra**), illegal boulder mining in the Ganges and Shivalik *raos*, and siltation of Jhilmil Jheel threaten habitat continuity.

7. Nainital District (Kumaon Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Nainital district (4,251 sq km) leads all 13 districts of Uttarakhand in proportional forest cover at **71.62%** (3,044.49 sq km) and holds the state's largest expanse of Very Dense Forest (762.49 sq km) (ISFR District Data). Spanning the Terai-Bhabar plains (250 m at Haldwani/Ramnagar) to the Lesser Himalayan Naina Peak ridges (2,615 m), Nainital encompasses four statutory protected areas:

- **Jim Corbett National Park & Tiger Reserve (Eastern & Southern Sector):** Established in 1936 as Hailey National Park (India's first National Park) and launched as the inaugural **Project Tiger Reserve** on 1 April 1973, its headquarters at **Ramnagar** and its premier eastern safari and buffer zones—**Bijrani, Garjiya, Dhela, Jhirna (eastern sector), Sitabani, and Phato/Terai West**—lie within Nainital district (Corbett Tiger Reserve Overview).
- **Nandhaur Wildlife Sanctuary (Western & Central Sector, 269.96 sq km):** Notified in December 2012 within the Haldwani Forest Division (Nandhaur, Jaulasal, and Chorgaliya ranges) between the Gola and Sharda rivers.
- **Pawalgarh Conservation Reserve (58.25 sq km):** Notified in December 2012 in the Ramnagar Forest Division between the Kosi and Dabka rivers (incorporating the Sitabani eco-tourism zone and Kyari-Pawalgarh Bhabar corridor).
- **Naina Devi Himalayan Bird Conservation Reserve (111.91 sq km):** Notified in 2015 inside the Nainital Forest Division across the **Kilbury-Pangot-Vinayak-Kunjkharak** oak-rhododendron ridge (1,400–2,615 m), complemented by the **Pt. G.B. Pant High Altitude Zoo (Nainital, 2,100 m)**.

- **Forest Cover & Types:** Managed by six major forest divisions—Nainital FD, Ramnagar FD, Haldwani FD, Terai Central FD, Terai West FD, and Corbett Tiger Reserve Division—the district's 3,044.49 sq km of forest cover spans 762.49 sq km Very Dense, 1,690.00 sq km Moderately Dense, and 592.00 sq km Open Forest (Status of Wildlife Habitats in Ramnagar FD). Along the Bhabar piedmont of Ramnagar, Kaladhungi, Kotabagh, and Chorgaliya (300–900 m), Moist Bhabar-Doon Sal Forests (*Shorea robusta*, *Terminalia alata*, *Anogeissus latifolia*) and Haldu-Kusum riverine forests dominate. Between 900 and 1,600 meters (Jeolikote, Bhowali, Garam Pani), sub-tropical Chir pine grows. From 1,600 to 2,615 meters across Nainital, Pangot, Mukteshwar, Ramgarh, and Maheshkhan, dense Moist Temperate Broadleaf Forests of Banj oak (*Quercus leucotrichophora*), Tilonj oak (*Quercus floribunda*), Kharsu oak (*Quercus semecarpifolia* on Naina Peak), Himalayan cypress (*Cupressus torulosa*), Deodar, and Burans thrive.
- **Aquatic Ecosystems & Water Bodies:** Styled the **Lake District of India**, featuring a cluster of tectonic-karstic Kumaon lakes (Tal systems): **Naini Tal** (1,938 m), **Bhimtal** (1,370 m, largest lake in Kumaon), **Naukuchiatal** (1,220 m, deepest lake in Kumaon at ~40 m), **Sattal** (1,370 m, an interconnected cluster of seven forest-ringed freshwater lakes: Panna/Garud Tal, Nal-Damyanti Tal, Ram-Lakshman-Sita Tal, Sukha Tal, and Purna Tal), **Khurpatal** (1,635 m), **Malwatal** (on the Kalsa River), **Harishtal**, and **Lokhamtal**. Major river basins draining from the Lesser Himalayas through the Bhabar into the Terai include the **Kosi River** (with Kosi Barrage at Ramnagar), **Gola River** (with Gola Barrage at Kathgodam/Haldwani), **Nandhaur River**, **Dabka River**, **Baur River**, and **Bhakra River**.

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 7. Nainital District:



Mammal: Bengal Tiger (Panthera tigris tigris) — Jim Corbett National Park Bijrani/Dhela



Bird: Great Hornbill (Buceros bicornis) — Pawalgarh Conservation Reserve Sal canopy



Reptile: King Cobra (Ophiophagus hannah) — Kaladhungi and Bailparao Bhabar ravines



Fish: Golden Mahseer (Tor putitora) — Kosi River at Garjiya Devi sanctuary

- **Mammals:** The Ramnagar, Corbett (Bijrani-Dhela-Garjiya), Pawalgarh, Terai West (Phato), and Haldwani-Nandhaur forests support one of the highest tiger and elephant densities on Earth. Over 300 of Uttarakhand's 560 Bengal tigers (All India Tiger Estimation 2022) utilize Nainital's Corbett buffer, Ramnagar FD (~65+ tigers), Terai West, and Nandhaur WLS (~30+ tigers), with camera traps recording tigers up to 2,400 meters in Maheshkhan and Mukteshwar oak forests. Over 600 Indian elephants traverse the Kosi-Dabka-Gola-Nandhaur corridors. Co-predators and herbivores include the Indian leopard, sloth bear (Bhabar), Asiatic black bear (Kilbury, Mukteshwar, Bhatrojkhan), dhole (*Cuon alpinus*), fishing cat (*Prionailurus viverrinus*), sambar, chital, hog deer, barking deer, Himalayan goral (Kilbury cliffs), Himalayan serow

(Vinayak-Kunjkharak oak-ringal gorges), yellow-throated marten, red giant flying squirrel, Indian pangolin, and smooth-coated otter.

- **Birds (Avifauna):** Richest birding landscape in the Western Himalayas, hosting over **550 bird species** across two complementary zones: (1) **Temperate Oak Zone (Naina Devi Himalayan Bird Conservation Reserve, Pangot, Kilbury, Sattal, Maheshkhan):** Cheer pheasant, Koklass pheasant, Kalij pheasant, rufous-throated partridge, long-tailed broadbill (*Psarisomus dalhousiae*), red-billed leiothrix, Siberian rubythroat, spotted and white-crested laughingthrushes, and collared owlet; and (2) **Bhabar-Terai & Riverine Zone (Corbett, Pawalgarh, Kosi Barrage, Nandhaur, Sattal):** Great hornbill, Oriental pied hornbill, tawny fish owl (*Ketupa flavipes*), spot-bellied eagle-owl (*Bubo nipalensis*), collared falconet, red-headed and white-rumped vultures, Pallas's fish eagle, ibisbill (wintering on Kosi shingle beds at Garjiya/Ramnagar), wallcreeper, and great slaty woodpecker (*Mulleripicus pulverulentus*) in mature Sal stands of Pawalgarh and Nandhaur.
- **Reptiles & Amphibians:** Kaladhungi, Kotabagh, Chorgaliya, and Bailparao moist Bhabar forests represent India's northernmost high-density breeding stronghold of the King cobra (*Ophiophagus hannah*), nesting between 400 and 1,900 meters (up to Bhowali and Jeolikote). Other reptiles include Gharial (*Gavialis gangeticus*) and Mugger crocodile (*Crocodylus palustris*) in the Kosi and Ramganga rivers, Burmese python (*Python bivittatus*), Russell's viper, bamboo pit viper, Himalayan pit viper (Kilbury/Mukteshwar), tokay gecko (*Gekko gecko*), and Critically Endangered elongated tortoise (*Indotestudo elongata*) in Corbett/Ramnagar Sal forests.
- **Insects & Butterflies: Sattal, Jones Estate (Bhimtal), and Pangot** (home to the Butterfly Research Centre at Bhimtal) support over **250 butterfly species**—including the Golden Emperor (*Dilipa morgiana*), Siren, Great Windmill, Red Helen, Paris Peacock, Common Raven, Spangled Plushblue, and the giant Atlas moth (*Attacus atlas*).
- **Aquatic Species:** The Kosi River (especially the protected stretch at Garjiya Devi and Betalghat), Nandhaur River, and Bhimtal-Naukuchiatal lakes support thriving populations of Golden mahseer (*Tor putitora*), Copper mahseer, snow trout (*Schizothorax richardsonii*), and *Goonch* catfish (*Bagarius yarrelli*).

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** Connectivity between Jim Corbett National Park, Pawalgarh Conservation Reserve, and Nandhaur Wildlife Sanctuary relies on two wildlife corridors: the **Kosi River Corridor (Mohaan-Sundarkhal-Chilkhiya-Sitabani)** and the **Gola River Corridor (Lalkuan-Tanda-Chorgaliya)**. Within mature Sal (*Shorea robusta*) stands of Pawalgarh and Nandhaur, large cavity-bearing trees (>80 cm DBH) provide essential nesting micro-habitats for the Great hornbill and Great slaty woodpecker, while moist Bhabar leaf litter supports the elongated tortoise and king cobra nests.
- **Statutory Conservation Status:** Priority Schedule I species include the Bengal tiger (IUCN Endangered; CITES Appendix I), Indian elephant (IUCN Endangered; CITES Appendix I), Gharial (IUCN Critically Endangered; CITES Appendix I), Elongated tortoise (IUCN Critically Endangered; CITES Appendix II), King cobra (IUCN Vulnerable; CITES Appendix II), Great hornbill (IUCN Vulnerable; CITES Appendix I), and Cheer pheasant (IUCN Vulnerable; CITES Appendix I).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** The **Kosi Corridor (Ramnagar-Dhikuli-Mohaan)** and **Haldwani-Lalkuan-Chorgaliya-Kaladhungi** interfaces experience chronic human-tiger, human-leopard, and human-elephant conflict. Along NH-309 (Ramnagar to Mohaan), dispersing tigers occasionally ambush livestock and forest-produce collectors, while in Bhimtal, Bhowali, Mukteshwar, Ramgarh, and Okhalkanda hill blocks, leopards attack livestock and humans. In Kotabagh, Bailparao, and Chorgaliya, elephants and wild boars raid paddy, wheat, and orchards. Mitigation includes 24x7 camera-trap and thermal drone surveillance along the Kosi corridor, the **Ramnagar-Dhela Wildlife Rescue Centre**, solar fencing, and King Cobra rescue squads in Kaladhungi/Haldwani.
- **Environmental Threats:** Two major anthropogenic pressures threaten Nainital's ecosystems: (1) **Luxury resort walling and linear construction along the Kosi River corridor (Dhikuli-Garjiya-Mohaan)**, which blocks tigers and elephants from accessing the Kosi River from eastern Corbett; and (2) **industrial-scale River Bed Material (RBM) boulder, gravel, and sand mining in the Gola, Nandhaur, Dabka, and Kosi rivers**, involving thousands of dump trucks daily from October to May, which severs the **Gola Corridor** between Terai Central and Nandhaur WLS and destroys mahseer spawning substrates. In the hill lakes, rapid urbanization and degradation of **Sukhatal** (the principal recharge wetland for Naini Lake) threaten local hydrology.

8. Pauri Garhwal District (Garhwal Division)

1. Ecosystem Profiles & Geographical Features

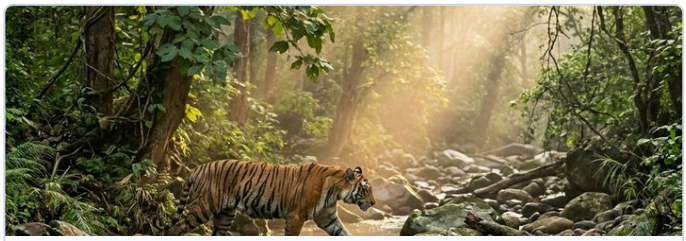
- **Protected Areas & Sanctuaries:** Encompassing 5,329 sq km from the southern Bhabar plains (295 m at Kotdwar) to the Dudhatoli-Lesser Himalayan crest (3,116 m), Pauri Garhwal district (official portal pauri.nic.in) holds the **largest absolute forest cover by area of any district in Uttarakhand at 3,396.71 sq km** (63.74% of geographical area) (ISFR District Ranking). It contains three major connected protected areas forming the **Rajaji-Lansdowne-Sonanadi-Corbett Tiger Megacorridor**:
  - **Jim Corbett National Park & Tiger Reserve (Western & Core Sector):** Nearly 60% of Corbett National Park's core area (312.76 sq km of the 520.82 sq km core)—including **Dhikala Chaur**, **Ramganga Reservoir (Kalagarh Dam)**, **Sarpduli**, **Sultan**, **Gairal**, **Khinnanauli**, and western **Jhirna/Kalagarh** zones—lies within Pauri Garhwal district under the Kalagarh Tiger Reserve Forest Division (headquartered at Lansdowne).
  - **Sonanadi Wildlife Sanctuary (301.18 sq km):** Established on 9 January 1987 in Pauri Garhwal (Wildlife Sanctuaries of Uttarakhand), forming the northwestern buffer zone of Corbett Tiger Reserve across the Sonanadi, Palain, and Mandal river valleys (Vatanvasa and Halduparao gates).
  - **Rajaji National Park & Tiger Reserve (Eastern / Gohri & Chilla Sector):** Encompasses the **Gohri Range** (Lakshman Jhula to Chilla) and eastern **Chilla Range** (~250 sq km) in the Yamkeshwar block of Pauri Garhwal (Pauri District Wildlife).
- **Forest Cover & Types:** Managed by Garhwal FD (Pauri), Lansdowne FD (Kotdwar/Dugadda), Kalagarh Tiger Reserve Division, Soil Conservation FD Lansdowne, and Civil Soyam FD Pauri, the district's 3,396.71 sq km of forest cover comprises 620.71 sq km Very Dense, 1,920.00 sq km Moderately Dense, and 856.00 sq km Open Forest (ISFR District Data). In the southern Kalagarh, Sonanadi, Kotdwar, Dugadda, and Laldhang Bhabar-Shivalik belt (300–1,100 m), Moist and Dry Siwalik Sal Forests (*Shorea robusta*), bamboo brakes (*Dendrocalamus strictus*), riverine Khair-Sissoo, and expansive alluvial grasslands (**Chaurs**, such as Dhikala, Phulai, Khinnanauli, and Paterpani Chaurs dominated by *Themeda arundinacea*, *Saccharum*, and *Vetiveria*) flourish. Across the middle hills of Lansdowne, Satpuli, Pokhra, Ekeshwar, and Pauri (1,100–2,000 m), Chir pine (*Pinus roxburghii*) and Banj oak (*Quercus leucotrichophora*) intergrade. In the high **Dudhatoli Range** ("Pamir of Uttarakhand", 2,200–3,116 m) and **Khirsu-Adwani-Binsar Mahadev** belt, Moist Temperate Forests of Kharsu and Moru oak, silver fir (*Abies pindrow*), spruce (*Picea smithiana*), yew (*Taxus wallichiana*), and Deodar surround high montane meadows (**Musagali, Kodeya, and Dudhatoli Bugyals**).
- **Aquatic Ecosystems & Water Bodies:** Bounded on the north and west by the **Alaknanda River** and **Ganges River** (formed at **Devprayag** at 472 m and flowing past Vyasghat, Kaudiyala, and Lakshman Jhula). Interior basins include two major systems originating in the **Dudhatoli crystalline massif**: (1) the **Nayar River**, formed by the confluence of the **Purvi (Eastern) Nayar** and **Paschimi (Western) Nayar** at **Satpuli (Banghat)**, which joins the Ganges at **Indraprayag (Vyasghat)**; and (2) the **Western Ramganga River**, which enters the southern Kalagarh gorge, receives the **Mandal**, **Palain**, and **Sonanadi** rivers, and forms the **80-sq-km Ramganga Reservoir (Kalagarh Dam)** inside Corbett National Park. Additional Shivalik streams include the **Khoh River** (Kotdwar), **Malin River** (Kanvashram), and **Rawasan River**, while **Tara Kund** (2,200 m) is a notable high-altitude tarn wetland.

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 8. Pauri Garhwal District:



Reptile: Gharial (Gavialis gangeticus) — Ramganga River sandbanks at Dhikala, Corbett NP



Mammal: Bengal Tiger (Panthera tigris tigris) — Kalagarh and Sonanadi Wildlife Sanctuary



Mammal: Asian Elephant (Elephas maximus) — Dhikala Chaur summer congregation



Fish: Golden Mahseer (Tor putitora) — Nayar River spawning runs at Satpuli & Banghat

- **Mammals:** The **Dhikala-Ramganga-Sonanadi-Lansdowne-Chilla** belt supports the highest biomass of wild ungulates and mega-carnivores in northern India. Over 150 of Corbett's 260 Bengal tigers occupy the Dhikala, Sarpduli, Kalagarh, Sonanadi, and adjoining Lansdowne FD (Kotdwar-Dugadda-Laldhang) ranges, while over 500 Indian elephants congregate seasonally on the **Dhikala Chaur** and Ramganga drawdown flats during summer (March–June). Across the middle hills (Pabau, Pokhra, Chaubattakhal, Rikhnikhal, Nainidanda, Khirsu, and Thalisain), the Indian leopard reaches some of the highest densities in Asia. Other mammals include sloth bear (Kalagarh/Lansdowne), Asiatic black bear (Dudhatoli and Khirsu), hog deer (*Axis porcinus*, abundant on Dhikala Chaur), sambar, chital, barking deer, Himalayan goral, Himalayan serow (Dudhatoli fir-oak forests), yellow-throated marten, smooth-coated otter (packs of 10–15 in the Ramganga, Palain, and Nayar rivers), and Indian pangolin.
- **Birds (Avifauna):** Over 580 bird species occur across the Dhikala-Ramganga wetland-grassland complex, Lansdowne hills, and Khirsu-Dudhatoli ridges. The Ramganga Reservoir and Dhikala Chaur support breeding Pallas's fish eagle (*Haliaeetus leucoryphus*), lesser fish eagle (*Icthyophaga humilis*), grey-headed fish eagle, osprey, cinereous, red-headed, and Himalayan griffon vultures, black-necked stork, wallcreeper, ibisbill, spot-bellied eagle-owl, and Great and Oriental pied hornbills. In Khirsu, Lansdowne, and Dudhatoli oak-deodar forests, Cheer pheasant, Koklass pheasant, Kalij pheasant, bearded vulture, and ultramarine flycatcher thrive.
- **Reptiles & Amphibians:** The **Ramganga Reservoir and River** inside Pauri's Dhikala-Gairal-Boxar stretch supports the **largest wild breeding population of the Critically Endangered Gharial (*Gavialis gangeticus*) in Uttarakhand** (>100 adults and basking cohorts on deep pools and sandbanks), alongside dense populations of Mugger crocodile (*Crocodylus palustris*). Terrestrial reptiles include the King cobra (*Ophiophagus hannah*, in Lansdowne, Dugadda, and Sonanadi bamboo-Sal gorges), Burmese python, Russell's viper, common krait, Himalayan pit viper (Dudhatoli), and elongated tortoise (*Indotestudo elongata*).
- **Insects & Butterflies:** The Mandal River valley (Rathuadhab-Kanda), Sonanadi, and Khirsu support over 180 butterfly species, including the Common Peacock, Blue Peacock (*Papilio arcturus*), Great Mormon, Red Lacewing (*Cethosia biblis*), Common Map, and Glassy Tiger, alongside *Apis dorsata* rock-bee colonies on the Ramganga gorge cliffs.
- **Aquatic Species:** The **Nayar River (at Banghat/Satpuli and Vyasghat)**, **Western Ramganga (Gairal-Marchula)**, **Sonanadi**, and **Palain** rivers constitute premier spawning grounds of the Golden mahseer (*Tor putitora*), where adult broodfish migrate upstream from the Ganges and Ramganga Reservoir during the monsoon, co-occurring with *Bagarius yarrelli* (*Goonch*), snow trout (*Schizothorax richardsonii*), and *Labeo dero*.

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** Two micro-habitats in Pauri Garhwal carry global conservation significance: (1) the **Dhikala and Khinnanauli Chaurs** (alluvial grasslands within the Sal matrix), which sustain dense aggregations of hog deer, chital, and Indian elephant, supporting Corbett's unmatched Bengal tiger density; and (2) the deep bedrock pools and sandbars of the **Ramganga River inside Dhikala-Boxar**, which provide undisturbed nesting substrates for the Gharial. Meanwhile, the **Lansdowne Forest Division (Laldhang-Kotdwar-Kalagarh)** serves as the indispensable genetic corridor between Rajaji and Corbett Tiger Reserves (Shivalik Hills HWC & Corridor Study).
- **Statutory Conservation Status:** Priority Schedule I species include the Gharial (IUCN Critically Endangered; CITES Appendix I), Bengal tiger (IUCN Endangered; CITES Appendix I), Indian elephant (IUCN Endangered; CITES Appendix I), Hog deer (IUCN Endangered; CITES Appendix I), Smooth-coated otter (IUCN Vulnerable; CITES Appendix I), and Indian pangolin (IUCN Endangered; CITES Appendix I).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Pauri Garhwal is consistently ranked as **Uttarakhand's #1 hotspot for human-leopard conflict** (Drivers and Socio-Ecological Consequences of HWC in Uttarakhand). Across Pauri, Pabau, Pokhra, Ekeshwar, Rikhnikhal, Bironkhal, Nainidanda, Khirsu, and Dugadda blocks, severe rural depopulation ("ghost villages") has allowed secondary scrub (*Lantana camara*, *Rubus*, and Chir pine) to engulf village courtyards, creating ambush cover for Indian leopards and resulting in recurrent human fatalities and livestock predation. In Rikhnikhal and Kaljikhal

(adjoining northern Corbett), dispersing **tigers** have also entered middle-hill villages. In the southern Kotdwar, Laldhang, and Yamkeshwar (Gohri) belt, elephants and wild boars inflict heavy crop damage (Lansdowne FD Conflict Study). Mitigation includes bio-fencing, Village Rapid Response Teams, GSM-enabled camera traps, and a dedicated Leopard Rescue Centre in Pauri.

- **Environmental Threats:** The **Kotdwar-Lansdowne-Laldhang wildlife corridor**—the sole link between Rajaji and Corbett Tiger Reserves—is threatened by road expansion along the **Laldhang-Chillarkhal route**, urban-industrial growth around Kotdwar and Saneh, and boulder mining in the Khoh, Malin, and Rawasan rivers. In Pauri's middle hills, recurring summer forest fires in the Chir pine belt destroy understory browse, while the **Srinagar Hydroelectric Dam (330 MW)** on the Alaknanda has altered sediment transport and blocked upstream mahseer migration above Srinagar.

9. Pithoragarh District (Kumaon Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Forming the easternmost frontier of Uttarakhand ("Little Kashmir"), Pithoragarh district (7,090 sq km; official portal pithoragarh.nic.in) is bordered by Tibet (China) to the north across the Lipulekh, Darma, and Unta Dhura passes, and by Nepal to the east across the Kali River. Its flagship protected area is the **Askot Musk Deer Sanctuary** **(Askot Wildlife Sanctuary)**, established in 1986 across **599.93 sq km** in the Dharchula and Munsiyari subdivisions (UKPSC Protected Areas). Spanning an altitudinal range from **600 meters** (at the Kali-Gori river valleys near Jauljibi/Askot) to **6,904 meters** (at **Panchachuli II**), Askot WLS sits at the junction of the **Kailash Sacred Landscape**—connecting the Western Himalayan and Central/Nepalese Himalayan biogeographic provinces (adjacent to Nepal's Api Nampa Conservation Area) and containing 111 Van Panchayats within its Eco-Sensitive Zone. Additionally, the northwestern **Milam-Martoli-Ralam** sector of Munsiyari borders the eastern buffer of the Nanda Devi Biosphere Reserve.
- **Forest Cover & Types:** Administered by Pithoragarh Forest Division (Pithoragarh, Didihat, Gangolihat, Berinag, Munakot, Dharchula, and Munsiyari ranges), the district supports 2,078.00 sq km of forest cover (29.35% of geographical area, with >65% of the district comprising alpine meadows, glaciers, and Trans-Himalayan cold desert above the timberline), including 517.60 sq km Very Dense, 975.47 sq km Moderately Dense, and 584.93 sq km Open Forest (ISFR 2023 Vol II). Because of its 6,300-meter vertical span, Pithoragarh exhibits every Himalayan vegetation type (Pithoragarh Official Flora & Fauna):
  - *Sub-tropical Riverine & Sal/Pine (600–1,600 m):* Moist Sal (*Shorea robusta*), Toon, Khair-Sissoo, and unique **Indian Butter Tree (*Diploknema butyracea*, locally *Chyura*)** forests along the warm, humid gorges of the Kali, Gori, and Saryu rivers (Jauljibi, Ghat, Thal), along with *Dendrocalamus* bamboo and Chir pine.
  - *Temperate Broadleaf & Conifer (1,600–3,000 m):* Extensive stands of **Rianj oak (*Quercus lanata*)** (at Narayan Ashram, Kalamuni, and Didihat—holding the largest *Q. lanata* area in Kumaon), Banj, Tilonj, and Kharsu oaks, *Rhododendron arboreum*, sacred **Deodar groves at Thal Kedar and Chandak**, and **Himalayan Hemlock (*Tsuga dumosa*)** and Taxus in the wet Gori-Darma gorges (Quercus lanata Kumaon Study).
  - *Sub-Alpine & Alpine Bugyals (3,000–4,800 m):* Birch (*Betula utilis*) and juniper scrub opening into vast alpine **Bugyals**: **Khaliya Top Bugyal** (3,500 m above Munsiyari), **Chiplakot Bugyal** (3,200–4,200 m), **Ralam Bugyal**, **Milam-Martoli**, **Darma-Panchachuli Base**, **Byans (Gunji-Kalapani-Nabhi)**, and **Chhota Kailash (Jollingkong)** (Pithoragarh Bugyal Tourism Regulation).
- **Aquatic Ecosystems & Water Bodies:** Carved by four immense trans-Himalayan and glacier-fed river valleys flowing north-to-south: (1) the **Kali (Mahakali/Sharda) River**, originating at **Kalapani** near Lipulekh Pass (3,600 m) and joined by the **Kuti Yangti** at Gunji; (2) the **Dhauliganga (Eastern) River**, draining the Darma and Lassar Yangti valleys and joining the Kali at **Tawaghat**; (3) the **Goriganga River**, originating from the **Milam Glacier** and Ralam Glacier in Munsiyari and joining the Kali at **Jauljibi Prayag**; and (4) the **Ramganga (Eastern) River**, originating from the **Namik and Ponting glaciers**, flowing through Birthi, Thal, and Nachani to join the **Saryu River** at **Rameshwar Prayag** (Ghat). High-altitude lakes include **Parvati Tal (Jollingkong Lake at 4,572 m below Adi Kailash)**, **Anchri Tal**, **Chhipla Kund**, **Thamri Kund** (near Khaliya Top, Munsiyari), and **Maheshwari Kund**.

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 9. Pithoragarh District:



Mammal: Alpine Musk Deer (Moschus chrysogaster) — Askot Musk Deer Sanctuary



Mammal: Snow Leopard (Panthera uncia) — Johar, Darma and Byans high valleys



Bird: Himalayan Monal (Lophophorus impejanus) — Khaliya Top and Kalamuni pass



Fish: Golden Mahseer (Tor putitora) — Mahakali River at Jauljibi & Jhulaghat

- **Mammals:** Askot Musk Deer Sanctuary and the Johar-Darma-Byans valleys represent one of the most taxonomically diverse mammal landscapes in the Himalaya. In WII camera-trap surveys in Askot WLS, a **Bengal tiger (*Panthera tigris tigris*) was photographed at 3,630 meters in snow**, overlapping directly with the habitat of the **Snow leopard (*Panthera uncia*)**, **Indian leopard (*Panthera pardus fusca*)**, and **Clouded leopard (*Neofelis nebulosa*)**. The upper Johar (Milam-Lwan), Darma, and Byans (Kuti) valleys support a high-density snow leopard population, Tibetan wolf (*Canis lupus chanco*), Eurasian lynx (*Lynx lynx isabellinus*), dhole (*Cuon alpinus*), Himalayan brown bear, and Asiatic black bear. Herbivores include the flagship **Alpine musk deer (*Moschus chrysogaster*)** and Himalayan musk deer (*Moschus leucogaster*) at Khaliya Top, Chiplakot, and Ralam (2,800–4,200 m), large herds of Bharal (*Pseudois nayaur*), Himalayan tahr (*Hemitragus jemlahicus*), Himalayan serow, Himalayan goral, Tibetan argali (*Ovis ammon*), Kiang (*Equus kiang*), sambar, barking deer, yellow-throated marten, stone marten (*Martes foina*), woolly flying squirrel, and Royle's and large-eared pikas (*Ochotona macrotis*).
- **Birds (Avifauna):** Munsiyari (Khaliya Top, Kalamuni Pass, Thamri Kund) and the Goriganga valley (Baram-Balgad) host over **350 bird species**, combining Western and Central/Eastern Himalayan elements. Key Galliformes include the State Bird Himalayan monal, **Satyr tragopan (*Tragopan satyra*, abundant at Kalamuni-Khaliya Top)**, Koklass pheasant, Cheer pheasant (steep grassy slopes of Askot and Thal Kedar), Blood pheasant (*Ithaginis cruentus*, easternmost Kali-Tinker border), Tibetan snowcock (*Tetraogallus tibetanus*), Himalayan snowcock, and snow partridge. Other avian highlights include the Bearded vulture, golden eagle, wren-babblers (*Pnoepyga* and *Spelaeornis*), cutia (*Cutia nipalensis*), fire-tailed sunbird, Gould's shortwing, wallcreeper, and spot-winged rosefinch.
- **Reptiles & Amphibians:** The warm, humid Goriganga and Kali gorges (Baram, Jauljibi, Dharchula) up to 2,200 meters harbor the King cobra (*Ophiophagus hannah*), green pit viper (*Trimeresurus septentrionalis*), Himalayan pit viper (*Gloydius himalayanus*), Mountain keelback (*Herpetoreas platyceps*), Asian glass lizard (*Dopasia gracilis*), Sikimmese skink, and high-altitude torrent amphibians (*Nanorana liebigii*, *Nanorana vicina*, and *Amolops formosus*).
- **Insects & Butterflies:** The **Goriganga Valley (Baram-Madkot-Munsiyari)** and **Didihat-Askot** are world-renowned butterfly hotspots supporting over **280 butterfly species**, including the **Kaiser-i-Hind (*Teinopalpus imperialis*, WPA Schedule I, in high oak canopies of Munsiyari/Didihat)**, Common Peacock, Blue Peacock, Krishna Peacock (*Papilio krishna*), Golden Emperor, Pasha (*Herona marathus*), and Regal and Common Red Apollos (*Parnassius* spp.) on Milam and Ralam bugyals.
- **Aquatic Species:** The Kali River (at Jauljibi, Jhulaghat, and Pancheshwar), Goriganga, Saryu, and Eastern Ramganga support major populations of Golden mahseer (*Tor putitora*), Chocolate/Copper mahseer (*Neolissochilus hexagonolepis*), snow trout (*Schizothorax richardsonii* and *Schizothorax kumaonensis*), *Bagarius yarrelli*, and specialized hillstream catfishes (*Myersglanis blythii* and *Euchiloglanis*).

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** The **Goriganga and Kali river gorges** act as deep thermal corridors allowing tropical/sub-tropical fauna—including the Bengal tiger, King cobra, and Oriental butterflies—to penetrate deep into the Greater Himalayan massif up to 3,600 meters, resulting in direct guild sympatry with the Snow leopard and Alpine musk

deer in Askot Musk Deer Sanctuary. Across the **Kailash Sacred Landscape**, transboundary ridges between Dharchula-Byans and Nepal's Api Nampa Conservation Area enable genetic connectivity for tahr, serow, and musk deer.

- **Statutory Conservation Status:** Priority Schedule I species include the Alpine musk deer (IUCN Endangered; CITES Appendix I), Snow leopard (IUCN Vulnerable; CITES Appendix I), Bengal tiger (IUCN Endangered; CITES Appendix I), Tibetan argali (IUCN Near Threatened; CITES Appendix I), Himalayan serow (IUCN Vulnerable; CITES Appendix I), and **Kaiser-i-Hind butterfly** (*Teinopalpus imperialis*, IUCN Near Threatened; CITES Appendix II; WPA Schedule I).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Severe human-leopard conflict occurs across Pithoragarh town fringes (Chandak, Sore Valley), Didihat, Kanalichina, Berinag, Gangolihat, and Munakot, where Indian leopards attack women collecting fodder and domestic livestock. In Munsiyari, Dharchula, Darma, and Johar valleys, Asiatic black bears (*Ursus thibetanus*) raid maize, potato, and kidney-bean (*Rajma*) crops and attack villagers, while snow leopards and Tibetan wolves depredate migratory Bhotiya sheep and goat herds on alpine pastures.
- **Environmental Threats:** Pithoragarh faces acute pressures from three sources: (1) **Cross-border wildlife poaching and unregulated *Keeda Jadi* (*Ophiocordyceps sinensis*) harvesting**, which drives thousands of collectors into Chiplakot, Ralam, Laspa, and Panchachuli bugyals every May–June (prompting the Pithoragarh Forest Division to impose strict camping and grazing limits on Khaliya Top and alpine meadows) (Pithoragarh Alpine Meadow Regulations); (2) **Hydropower and strategic road blasting**, including the **Dhauliganga Dam (280 MW at Chirkila, Tawaghat)** and heavy blasting along the **Lipulekh Highway (Tawaghat-Ghatiabagarh-Gunji)** and **Munsiyari-Milam road**; and (3) **magnesite and soapstone mining** around Chandak and Jhirauli.

10. Rudraprayag District (Garhwal Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Established in 1997 from contiguous parts of Chamoli, Tehri, and Pauri, Rudraprayag district (1,984 sq km; official portal rudraprayag.gov.in) is the second-smallest district in Uttarakhand yet ranks among the state's highest in proportional forest cover (**57.58%** in ISFR 2021; **57.79%** in ISFR 2023) (ISFR District Ranking). The entire northern half of the district—stretching from Phata, Triyuginarayan, Sonprayag, and Gaurikund up to the **Kedarnath Dome (6,831 m), Kedarnath Main (6,940 m), Mandani (6,193 m), and Chaukhamba (7,068 m)** peaks, and eastward across the **Ukhimath-Sari-Chopta-Tungnath-Madhyamaheshwar** ridge—comprises the western core and buffer of **Kedarnath Wildlife Sanctuary** (975.20 sq km total, notified in 1972) (LotusArise Kedarnath WLS).
- **Forest Cover & Types:** Administered by Rudraprayag Forest Division and Kedarnath Wildlife Sanctuary FD (Guptkashi and Ukhimath ranges), the district's 1,142.30 sq km of forest cover includes 246.30 sq km Very Dense, 588.00 sq km Moderately Dense, and 308.00 sq km Open Forest (ISFR District Data). According to IUCN assessments of Kedarnath WLS, 44.4% to 48.8% of the sanctuary is forested, 7.7% comprises alpine meadows and krummholz scrub, and 42.1% is high-altitude rock, moraine, and permanent glacier ice (Kedarnath WLS Wikipedia). Sub-tropical Chir pine (*Pinus roxburghii*) clothes the lower Mandakini and Alaknanda valleys (600–1,600 m from Rudraprayag to Tilwara and Agastyamuni). Between 1,600 and 3,000 meters across Ukhimath, Guptkashi, Triyuginarayan, Ransi, and Chopta, Moist Temperate Broadleaf and Coniferous Forests of Banj, Moru, and Kharsu oak, horse chestnut, walnut, alder, silver fir (*Abies pindrow*), and multi-species rhododendrons (*Rhododendron arboreum*, *R. barbatum*, *R. campanulatum*) occur, underlain by dense Ringal bamboo. Above 3,000 meters, sub-alpine birch (*Betula utilis*) opens into iconic alpine **Bugyals**: **Chopta-Tungnath-Chandrashila Bugyal** (2,900–4,000 m), **Madhyamaheshwar and Kachni Bugyal** (3,497–4,200 m), **Mandani Bugyal**, **Bisuri Dhar**, and **Kham Bugyal**.
- **Aquatic Ecosystems & Water Bodies:** Anchored by the **Mandakini River**, which originates from the **Chorabari Glacier** (3,895 m) above Kedarnath and flows 72 km south to join the **Alaknanda River** at the sacred confluence of **Rudraprayag** (610 m). Major tributaries include the **Vasukiganga (Son Ganga) River** (originating from Vasuki Tal and joining the Mandakini at **Sonprayag**, 1,829 m), **Madhyamaheshwar Ganga** and **Markandeya Ganga** (joining at **Kalimath-Guptkashi**), **Khir Ganga**, and **Lastar Gad** (joining at Tilwara). High-altitude lakes include **Deoria Tal** (2,438 m above Sari village), **Vasuki Tal** (4,135 m west of Kedarnath), **Chorabari Tal (Gandhi Sarovar)** (3,900 m), and **Badhani Tal** (Jakholi block).

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 10. Rudraprayag District:



Bird: Himalayan Monal (Lophophorus impejanus) — Chopta-Tungnath in Kedarnath WLS



Mammal: Himalayan Tahr (Hemitragus jemlahicus) — Tungnath & Chandrashila rocky crags



Mammal: Alpine Musk Deer (Moschus chrysogaster) — Madhyamaheshwar sub-alpine birch scrub



Mammal: Indian Leopard (Panthera pardus fusca) — Middle-hill forests of Guptkashi & Ukhimath

- **Mammals:** The upper Mandakini, Vasuki Tal, Mandani, and Madhyamaheshwar basins (3,200–5,200 m) support resident Snow leopard (*Panthera uncia*) and Himalayan brown bear (*Ursus arctos isabellinus*), while the temperate and sub-tropical belts harbor dense populations of Indian leopard and Asiatic black bear (*Ursus thibetanus*). In camera-trap surveys, a **Bengal tiger (*Panthera tigris tigris*) was photographed at 3,274 meters in Kedarnath Wildlife Sanctuary**, proving active upward dispersal along the Alaknanda-Mandakini corridor. Ungulate diversity includes **Alpine musk deer (*Moschus chrysogaster*)** in the Tungnath-Chandrashila, Madhyamaheshwar, and Kedarnath birch-rhododendron zones (2,800–4,000 m), large herds of **Himalayan tahr (*Hemitragus jemlahicus*)** on the steep rocky crags of Tungnath, Ransi, and Gaurikund (2,200–4,200 m), Bharal (*Pseudois nayaur*) around Vasuki Tal, Himalayan serow in moist Ringal ravines of Duggalbitta and Makku, Himalayan goral, sambar, barking deer, Royle's pika (*Ochotona roylei*—abundant in rock talus along the Tungnath and Kedarnath trails), yellow-throated marten, red fox, and red giant flying squirrel (Sari-Deoria Tal).
- **Birds (Avifauna):** The **Sari–Deoria Tal–Duggalbitta–Chopta–Tungnath** altitudinal transect is one of the highest-density viewing sites on Earth for the State Bird **Himalayan monal (*Lophophorus impejanus*)**, alongside Koklass pheasant (*Pucrasia macrolopha*), Satyr tragopan (*Tragopan satyra*), Kalij pheasant, snow partridge (*Lerwa lerwa* at Chandrashila), and Himalayan snowcock. Raptors and high-altitude passerines include the Bearded vulture (nesting on Chandrashila and Gaurikund cliffs), golden eagle, Himalayan griffon, grandala, pink-browed and dark-breasted rosefinches, golden bush robin, fire-tailed sunbird, speckled wood pigeon, and white-capped and plumbeous water redstarts along the Mandakini.
- **Reptiles & Amphibians:** Includes the Himalayan pit viper (*Gloydius himalayanus*) in rocky rhododendron-birch zones up to 3,400 meters, Himalayan keelback (*Herpetoreas platyceps*), common rat snake, Himalayan agama (*Paralaudakia himalayana*) on slate boulders from Sonprayag to Rambara, Himalayan ground skink (*Asymblepharus himalayanus*), and cold-torrent amphibians (*Nanorana liebigii*, *Nanorana minica*, and *Duttaphrynus himalayanus*) in Deoria Tal and Makku streams.
- **Insects & Butterflies:** Deoria Tal, Makku Math, and Chopta meadows support Common Blue Apollo (*Parnassius hardwickii*), Common Peacock (*Papilio bianor*), Yellow Coster, Indian Tortoiseshell (*Aglais caschmirensis*), Painted Lady, Queen of Spain Fritillary, and high-altitude *Bombus* bumblebees pollinating alpine *Potentilla*, *Anemone*, and *Rhododendron* blooms.
- **Aquatic Species:** The Mandakini, Vasukiganga, Madhyamaheshwar Ganga, and Lastar Gad support cold-water snow trout (*Schizothorax richardsonii*, *Schizothorax plagiostomus*) and hillstream catfishes (*Pseudecheneis sulcata* and *Glyptothorax* spp.), while the Alaknanda-Mandakini confluence at Rudraprayag historically supported migratory Golden mahseer (*Tor putitora*) and *Goonch* (*Bagarius yarrelli*).

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** The **Chopta–Tungnath–Madhyamaheshwar ridge** (2,600–4,000 m) demonstrates classic vertical niche partitioning among four sympatric Himalayan ungulates: Alpine musk deer solitarily inhabit dense sub-alpine *Rhododendron campanulatum–Betula utilis* krummholz; Himalayan tahr occupy sheer south-

facing rocky cliffs and grassy ledges; Himalayan serow shelter in dark, moist temperate *Quercus–Abies–Ringal* gullies (2,000–3,000 m); and Himalayan goral graze lower rocky pine-oak slopes (1,000–2,200 m).

- **Statutory Conservation Status:** Priority Schedule I species include the Alpine musk deer (IUCN Endangered; CITES Appendix I), Snow leopard (IUCN Vulnerable; CITES Appendix I), Himalayan tahr (IUCN Near Threatened), Himalayan serow (IUCN Vulnerable; CITES Appendix I), Himalayan monal (CITES Appendix I), and Bearded vulture (IUCN Near Threatened; CITES Appendix II).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Following the historical legacy of the **Man-Eating Leopard of Rudraprayag** (shot by Jim Corbett at Golabrai in 1926), Rudraprayag remains a chronic hotspot for human-leopard and human-bear conflict across Augustmuni, Jakholi, Tilwara, Guptkashi, and Ukhimath blocks. Leopards stalk livestock and attack villagers along terraced fields near Rudraprayag town, Ratura, and Jakholi, while Asiatic black bears raid maize and amaranth crops in Ransi, Gaundar, Triyuginarayan, and Makku villages.
- **Environmental Threats:** The **Kedarnath Valley inside and bordering Kedarnath Wildlife Sanctuary** faces intense pilgrimage and infrastructure pressures: (1) **Helicopter & Pilgrimage Pressure:** Over 2 million pilgrims visit Kedarnath between May and October, accompanied by hundreds of daily low-altitude helicopter sorties from Phata, Sersi, and Guptkashi flying through the narrow Mandakini gorge at >85–95 dB, disrupting foraging and breeding of musk deer, tahr, monal, and bearded vultures; (2) **Hydropower & Linear Infrastructure:** The **Singoli-Bhatwari (99 MW)** and **Phata-Byung (76 MW)** hydroelectric projects on the Mandakini, combined with slope cutting for the **Char Dham Highway (NH-107)** and the **Rishikesh-Karnaprayag Railway Project**, fragment riparian habitats; and (3) **Unregulated Tourism in Chopta:** Proliferation of tented camps and solid waste along the Duggalbitta-Baniyakund-Chopta stretch degrades sub-alpine forest margins.

11. Tehri Garhwal District (Garhwal Division)

1. Ecosystem Profiles & Geographical Features

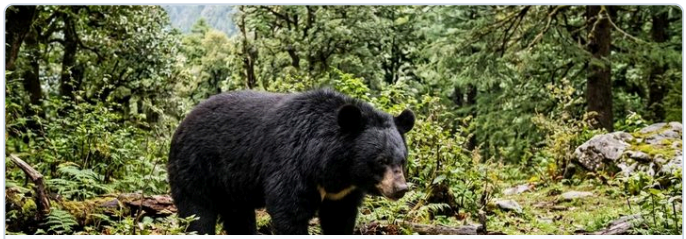
- **Protected Areas & Sanctuaries:** Covering 3,642 sq km between the Bhagirathi and Alaknanda-Bhilangana valleys, Tehri Garhwal district (official portal tehri.nic.in) extends from the southern Shivalik-Ganga gorge at Muni-ki-Reti/Tapovan (340 m) to the glaciated **Khatling-Sahasratal massif** (6,578 m). Its southwestern edge near Muni-ki-Reti/Narendra Nagar abuts the northern buffer of **Rajaji National Park**, while its northeastern **Bhilangana-Khatling** crest directly adjoins the western boundary of **Kedarnath Wildlife Sanctuary**. The district also encompasses the **Devalsari Community Biodiversity & Butterfly Sanctuary** (in the Aglar valley near Thatyur, Jaunpur block, adjoining Mussoorie WLS) and the **Bhadraj–Dhanaulti–Kaudia Eco-Forest Reserve**.
- **Forest Cover & Types:** Managed by Tehri Forest Division, Narendra Nagar FD, and Tehri Dam I & II Forest Divisions, the district maintains **2,064.39 sq km of forest cover** (56.68% of geographical area in ISFR 2021; rising to **61.03%** in ISFR 2023 to overtake Rudraprayag among mid-hill Garhwal districts), comprising 356.39 sq km Very Dense, 1,180.00 sq km Moderately Dense, and 528.00 sq km Open Forest (ISFR District Ranking). In the southern Narendra Nagar, Shivpuri, Byasi, and Kaudiyala Ganga gorge (350–1,100 m), Northern Tropical Dry and Moist Mixed Deciduous Sal-Anogeissus-Bamboo forests occur. Across the mid-altitudes of Chamba, New Tehri, Pratapnagar, Devprayag, and Ghansali (1,000–1,900 m), extensive Chir pine (*Pinus roxburghii*) forests cover warm slopes. From 1,900 to 3,200 meters across **Dhanaulti, Kanatal, Surkanda Devi (2,756 m), Chandrabadni, Devalsari (1,800–2,400 m), and Pinswar/Gangi**, dense Moist Temperate Banj, Moru, and Kharsu oak, Burans, horse chestnut, and **Himalayan Deodar (*Cedrus deodara*)** and silver fir forests thrive. Above 3,300 meters in the upper Bhilangana basin, sub-alpine birch-rhododendron transitions into spectacular alpine **Bugyals**: **Khatling Bugyal**, **Panwali Kantha Bugyal** (3,300–3,963 m), **Kush Kalyan Bugyal**, and **Kyarki Bugyal**.
- **Aquatic Ecosystems & Water Bodies:** Dominated by the **Bhagirathi River** and its principal tributary, the **Bhilangana River** (originating from the **Khatling Glacier** at 3,717 m). Their historic confluence at Old Tehri (Ganesh Prayag) is now submerged beneath the **42-sq-km Tehri Dam Reservoir (Swami Ram Tirath Sagar)**—India's highest dam (260.5 m) and Uttarakhand's largest artificial aquatic ecosystem, backed by the downstream **Koteshwar Dam Reservoir**. At the southeastern apex of the district lies **Devprayag** (472 m), where the Bhagirathi and Alaknanda merge to form the **Ganges River**. In the western Jaunpur block, the **Aglar River** and **Palar Gad** drain into the Yamuna, while the **Hiyunl (Henwal) River** drains the Chamba-Narendra Nagar valley into the Ganges at Phoolchatti. High-altitude glacial lakes include the seven-lake cluster of **Sahasratal** (4,572 m), **Masar Tal** (3,675 m), **Mahasar Tal**, **Yama Tal**, and **Apsara Tal** near Gangi/Khatling.

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 11. Tehri Garhwal District:



Mammal: Himalayan Serow (Capricornis sumatraensis thar) — Devalsari & Dhanaulti oak ravines



Mammal: Himalayan Black Bear (Ursus thibetanus laniger) — Upper Bhilangana & Gangi



Bird: Himalayan Monal (Lophophorus impejanus) — Surkanda Devi & Panwali Kantha Bugyal



Fish: Golden Mahseer (Tor putitora) — Ganges lotic stretch below Devprayag

- **Mammals:** As documented by the Tehri Garhwal Official Fauna Portal, apex predators in the high-altitude Khatling, Sahasratal, and Panwali Kantha basins (3,400–5,200 m) include the Snow leopard (*Panthera uncia*) and Himalayan brown bear (*Ursus arctos isabellinus*), while the Indian leopard (*Panthera pardus fusca*) and Asiatic black bear (*Ursus thibetanus*) are abundant throughout Tehri, Pratapnagar, Ghansali, Chamba, and Narendra Nagar forest divisions. In the southern **Narendra Nagar FD (Shivpuri-Byasi-Muni-ki-Reti)** adjacent to Rajaji Tiger Reserve, Bengal tigers and Indian elephants occasionally enter from the Dehradun-Haridwar-Pauri Terai/Shivalik belt (Tehri District Fauna). Ungulates include Bharal (*Pseudois nayaur*) and Himalayan tahr around Khatling and Sahasratal, Alpine musk deer on Panwali Kantha and Kush Kalyan ridges, Himalayan serow in dense oak-deodar ravines of Dhanaulti, Surkanda, and Devalsari, Himalayan goral on steep Bhagirathi and Ganga cliffs, sambar, barking deer, wild boar, yellow-throated marten, red giant flying squirrel (abundant in Devalsari and Dhanaulti deodar canopies), and smooth-coated otter in the Henwal and lower Ganga tributaries.
- **Birds (Avifauna):** The **Devalsari–Magra–Dhanaulti–Surkanda** belt and **Kaudia Forest** near Kanatal host over **260 avian species**. Key residents and migrants include the State Bird Himalayan monal (descending to Surkanda Devi, Panwali Kantha, and Pinswar in winter), Cheer pheasant (steep grassy slopes of Jaunpur, Nag Tibba base, and Chamba), Koklass pheasant, Kalij pheasant, Bearded vulture, Himalayan griffon, steppes eagle, green and purple cochoas (in moist Devalsari ravines), spot-winged starling, ultramarine flycatcher, and winter waterbird congregations (bar-headed geese, ruddy shelducks, great cormorants, ospreys, and Pallas's gulls) on the **42-sq-km Tehri Reservoir**.
- **Reptiles & Amphibians:** Includes the King cobra (*Ophiophagus hannah*) and Russell's viper in the lower Shivpuri-Narendra Nagar-Hiyunl valleys, Himalayan pit viper (*Gloydius himalayanus*) around Dhanaulti, Nag Tibba, and Ghansali, common krait, Oriental rat snake, Himalayan agama (*Paralaudakia himalayana*), Kashmir rock agama, and torrent amphibians (*Nanorana liebigii* and *Amolops* spp.).
- **Insects & Butterflies: Devalsari (Thatyur, Jaunpur block)** is celebrated as one of India's premier community-led butterfly sanctuaries, with over **170 documented butterfly species** in a single valley—including the rare **Spangled Plushblue (*Flos asoka*)**, **Pasha (*Herona marathus*)**, **Siren (*Hestina persimilis*)**, **Blue Admiral**, **Common Peacock**, **Great Windmill**, **White Admiral**, and numerous *Zephyrus* hairstreaks (*Euaspa*, *Chrysozephyrus*) obligatory on Banj and Moru oaks.
- **Aquatic Species:** While the 42-sq-km Tehri Dam reservoir supports lacustrine populations of Golden mahseer (*Tor putitora*), snow trout, and introduced carps, free-flowing upstream reaches of the **Bhilangana River** (above Ghansali), the **Aglar River** (site of the historical community fishing festival *Maun Mela*), and the **Ganges below Devprayag (Kaudiyala-Shivpuri)** remain vital lotic habitats for *Tor putitora*, *Schizothorax richardsonii*, *Barilius*, and *Garra* species.

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** The **Panwali Kantha–Khatling–Sahasratal** high-altitude corridor (3,300–4,800 m) provides crucial east-west connectivity between the Gangotri National Park landscape in Uttarkashi and the Kedarnath Wildlife Sanctuary landscape in Rudraprayag/Chamoli, enabling gene flow for Snow leopard, Alpine musk

deer, and Himalayan tahr. At mid-elevations (1,800–2,600 m), old-growth Deodar-Oak stands at Devalsari and Dhanaulti provide essential canopy micro-habitats for monophagous hairstreak butterflies and flying squirrels.

- **Statutory Conservation Status:** Priority Schedule I species include the Snow leopard (IUCN Vulnerable; CITES Appendix I), Alpine musk deer (IUCN Endangered; CITES Appendix I), Indian leopard (IUCN Vulnerable; CITES Appendix I), Himalayan serow (IUCN Vulnerable; CITES Appendix I), Cheer pheasant (IUCN Vulnerable; CITES Appendix I), and Golden mahseer (IUCN Endangered).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Empirical studies identify Tehri Garhwal district as one of the three most severe human-wildlife conflict districts in Uttarakhand (Impact of HWC in Tehri District). Across Bhilangana, Pratapnagar, Jakhnidhar, Chamba, Thauldhar, Devprayag, Kirtinagar, and Narendra Nagar blocks, Indian leopard attacks on humans and livestock depredation are pervasive. Concurrently, wild boars (*Sus scrofa*), rhesus macaques (*Macaca mulatta*), porcupines (*Hystrix indica*), and barking deer cause 40–60% crop losses across rainfed hill terraces. In the upper Bhilangana (Gangi, Pinswar), Asiatic black bears attack livestock and villagers.
- **Environmental Threats:** Impoundment of the **Tehri Dam (2,400 MW)** and **Koteshwar Dam (400 MW)** submerged 4,200 hectares of river valley and riparian forest, permanently blocking the upstream migratory pathway of Golden mahseer from the Ganges at Devprayag into the upper Bhagirathi and Bhilangana rivers, while fluctuating reservoir levels trigger drawdown-zone landslides around Pratapnagar and Chinyalisaur. Widening of the **Rishikesh-Chamba-Dharasu All-Weather Highway (NH-34)**, summer Chir pine forest fires, and commercial white-water rafting camps along the **Kaudiyala-Byasi-Shivpuri** Ganga stretch disturb nocturnal wildlife movement along the northern edge of Rajaji Tiger Reserve.

12. Udham Singh Nagar District (Kumaon Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Carved out of southern Nainital in October 1995 and spanning 2,542 sq km across the flat alluvial plains south of the Bhabar, Udham Singh Nagar district (headquartered at Rudrapur) represents the classic **Terai ecosystem** of Uttarakhand (elevation 190–250 m), where underground streams disappearing beneath the porous boulder-gravel Bhabar re-emerge along a high-water-table spring line (Terai-Bhabar Socio-Ecological Study). While the district has no standalone National Park within its revenue boundary, its eastern **Surai, Khatima, and Kilpura forest ranges** (Terai East Forest Division) constitute the internationally vital **Surai–Khatima–Kilpura–Boom Transboundary Wildlife Corridor**, directly linking **Nandhaur Wildlife Sanctuary** to the north with **Pilibhit Tiger Reserve (Uttar Pradesh)** to the south and **Shuklaphanta National Park (Nepal)** to the east (Surai-Khatima-Kilpura Corridor Study). The **Surai Ecotourism Range** (Khatima) serves as Uttarakhand's first dedicated Terai jungle safari zone outside a National Park, alongside the **Kakra Crocodile Trail** on Kakra Nala.
- **Forest Cover & Types:** Because over 70% of the fertile marshy Terai was reclaimed for intensive paddy, wheat, and sugarcane agriculture after 1947 and industrial estates at SIDCUL Pantnagar, Rudrapur, Kashipur, and Sitarganj, Udham Singh Nagar records the **lowest proportional forest cover of any district in Uttarakhand at 16.84%** (428.00 sq km: 121.00 sq km Very Dense, 205.00 sq km Moderately Dense, and 102.00 sq km Open Forest) (ISFR District Data). These remaining Terai forests are managed across three territorial divisions—**Terai East Forest Division** (Surai, Khatima, Kilpura, Ransali, and Doli ranges), **Terai Central Forest Division** (Tanda, Pipalparao, Barhani, and Gadarpur ranges), and **Terai West Forest Division** (Jaspur, Kashipur, and Ampokhra ranges). Forest types include Moist Terai Sal Forests (*Shorea robusta*, Type 3C/C2c, in **Surai and Kilpura ranges**), Low Alluvial Savannah Woodland (*Bombax ceiba–Albizia procera*), marshy *Syzygium cumini (Jamun)–Trewia nudiflora (Gutel)* swamp gallery forests, wet tall Terai *Phragmites–Saccharum–Vetiveria* grasslands, and commercial agro-forestry plantations of Eucalyptus, Poplar (*Populus deltoides*), and Teak (*Tectona grandis*) (Terai-Bhabar Comparative Forest Study).
- **Aquatic Ecosystems & Water Bodies:** Possesses the densest network of large man-made **Terai wetland reservoirs (Sagar/Dam systems)** in Uttarakhand (Wetlands of Uttarakhand Technical Report):
  - **Nanak Sagar Reservoir** (on the **Deoha / Nandhaur River** near Sitarganj/Nanakmatta, ~46 sq km water spread);
  - **Baigul (Sukhi) Reservoir** (near Sitarganj/Kichha);
  - **Dhora Reservoir** (near Kichha);
  - **Haripura and Baur Reservoirs** (on the **Baur and Bhakra rivers** near Bazpur/Gadarpur); and
  - **Tumaria (Tumariya) Reservoir** (on the **Phika and Dhela rivers** near Kashipur/Jaspur).

Major rivers traversing the district include the **Sharda (Mahakali) River** and **Sharda Canal** in the east (Khatima), **Deoha (Nandhaur)**, **Kamin**, **Gola (Kichha)**, **Bhakra**, **Baur**, **Kosi**, **Dhela**, and **Phika** rivers, alongside **Kakra Nala** (Surai Range)

and **Drona Sagar** (Kashipur).

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 12. Udham Singh Nagar District:



Mammal: Northern Swamp Deer / Barasingha (Rucervus duvaucelii) — Terai East & Central reedbeds



Mammal: Bengal Tiger (Panthera tigris tigris) — Surai Ecotourism Zone & Kilpura corridor



Mammal: Asian Elephant (Elephas maximus) — Kilpura-Khatima-Surai transboundary corridor



Fish: Golden Mahseer (Tor putitora) — Sharda Canal headworks at Lohiahead

- **Mammals:** Despite agricultural conversion, Udham Singh Nagar's three Terai forest divisions—especially **Surai-Kilpura-Ransali (Terai East)**, **Tanda-Pipalparao (Terai Central)**, and **Jaspur-Ampokhra (Terai West)**—support over **70+ resident Bengal tigers (*Panthera tigris tigris*)** (including "sugarcane tigers" breeding in tall cane fields bordering Surai, Pilibhit, and Tanda), alongside seasonal herds of Indian elephant (moving along the Kilpura-Khatima-Surai and Tanda corridors), Indian leopard, and sloth bear (*Melursus ursinus*). Critically, the wet grasslands and reservoir fringes of Terai East and Terai Central harbor the **Northern Swamp Deer / Barasingha (*Rucervus duvaucelii*)**, **Hog deer (*Axis porcinus*)**, sambar, chital, nilgai, barking deer, wild boar, **Fishing cat (*Prionailurus viverrinus*)**, leopard cat, rusty-spotted cat, smooth-coated otter (*Lutrogale perspicillata*, in Nanak Sagar, Haripura-Baur, and Sharda canals), honey badger, and historical/transient records of the **Hispid hare (*Caprolagus hispidus*)** along the Surai-Pilibhit-Shuklaphanta border.
- **Birds (Avifauna):** The five large Terai reservoirs (**Tumaria, Haripura-Baur, Nanak Sagar, Baigul, and Dhora**) and Pantnagar wetlands attract **tens of thousands of migratory Palearctic waterfowl** every winter and support over **350 avian species**. Key threatened and wetland-specialist species include the **Sarus crane (*Antigone antigone*, breeding in paddy-wetland mosaics of Khatima, Sitarganj, and Gadarpur)**, **Finn's weaver (*Ploceus megarhynchus*, Endangered, nesting in *Bombax ceiba* and reedbeds around Haripura-Baur and Gadarpur)**, Swamp francolin (*Ortygornis gularis*), Bengal florican (*Houbaropsis bengalensis*, historical Terai grassland habitat in Surai-Pilibhit border), black-necked stork, lesser adjutant, painted stork, Oriental darter, bar-headed goose, falcated duck (*Mareca falcata*), ferruginous pochard, osprey, Pallas's fish eagle, and Great and Oriental pied hornbills in Surai-Kilpura Sal stands.
- **Reptiles & Amphibians:** A baseline study by the Wildlife Institute of India in the Surai–Khatima–Kilpura corridor documented a rich Terai herpetofaunal guild: the **Mugger crocodile (*Crocodylus palustris*)**, which reaches its highest concentration in Uttarakhand in **Kakra Nala (Surai Range, >100 crocodiles)**, **Baur-Haripura Reservoir**, **Nanak Sagar**, and **Dhora-Baigul reservoirs** (Mugger Crocodile Distribution in TAL), alongside dispersing Gharial in the Sharda and Kosi systems. Snakes and chelonians include the **King cobra**, **Burmese python (*Python bivittatus*)**, banded krait (*Bungarus fasciatus*), common krait, monocled cobra (*Naja kaouthia*), spectacled cobra, Russell's viper, Siebold's smooth water snake (*Ferania sieboldii*), Indian eyed turtle (*Morenia petersi*), crowned river turtle (*Hardella thurjii*), three-striped roofed turtle, Indian softshell turtle, and yellow monitor (*Varanus flavescens*).
- **Insects & Butterflies:** Moist Sal and swamp corridors of Surai, Kilpura, and Pantnagar support lowland Oriental butterflies including the Blue Mormon (*Papilio polymnestor*), Lime Butterfly, Common Rose, Tawny Coster, Peacock Pansy (*Junonia almana*), Commander, Common Crow, and dense odonate guilds across the Baur and Nanak Sagar reedbeds.
- **Aquatic Species:** Terai reservoirs and lowland rivers (Sharda, Deoha, Baur, Kosi, Dhela) support rich warm-water Gangetic ichthyofauna: Rohu (*Labeo rohita*), Catla (*Labeo catla*), Mrigal (*Cirrhinus mrigala*), Calbasu (*Labeo calbasu*), Golden mahseer (*Tor putitora*, in the Sharda Canal headworks at Khatima-Lohiahead), Chitala featherback (*Chitala*

chitala), Bronze featherback (*Notopterus notopterus*), Giant river-catfish (*Sperata seenghala*), *Wallago attu*, stinging catfish (*Heteropneustes fossilis*), walking catfish (*Clarias magur*), and spiny eels (*Mastacembelus armatus*).

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** The **Surai–Khatima–Kilpura–Boom Corridor** is the sole remaining transboundary terrestrial link allowing Bengal tiger and Indian elephant exchange between the Indian Terai (Nandhaur WLS and Pilibhit Tiger Reserve) and Nepal's lowland Terai (Shuklaphanta National Park). Simultaneously, the seasonal drawdown mudflats, *Typha–Phragmites* reedbeds, and *Bombax* trees around **Haripura-Baur and Tumaria reservoirs** provide critical breeding micro-habitats for the globally Endangered **Finn's weaver (*Ploceus megarhynchus*)** and Vulnerable **Sarus crane (*Antigone antigone*)**.
- **Statutory Conservation Status:** Priority Schedule I species include the Bengal tiger (IUCN Endangered; CITES Appendix I), Indian elephant (IUCN Endangered; CITES Appendix I), Swamp deer / Barasingha (IUCN Vulnerable; CITES Appendix I), Hog deer (IUCN Endangered; CITES Appendix I), Fishing cat (IUCN Vulnerable; CITES Appendix II), Mugger crocodile (IUCN Vulnerable; CITES Appendix I), Finn's weaver (IUCN Endangered), and Sarus crane (IUCN Vulnerable; CITES Appendix II).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** Driven by the juxtaposition of dense forest blocks (Surai, Tanda, Pipalparao, Jaspur) with agricultural villages growing tall sugarcane and paddy. In **Khatima (Surai border), Sitarganj, Gadarpur, Bazpur, and Jaspur**, female tigers and leopards whelp cubs inside sugarcane fields, leading to close-range encounters during cane harvesting (November–March). Simultaneously, elephant herds moving through the **Kilpura-Khatima** and **Tanda-Lalkuan** corridors raid paddy and wheat crops, while **Mugger crocodiles** enter village fishponds and irrigation canals during monsoon floods. Mitigation includes the **Kakra Crocodile Rescue & Eco-Zone**, cane-harvester safety protocols, solar fencing, and elephant trenching.
- **Environmental Threats:** The **Kilpura–Khatima–Surai transboundary corridor** is pressured by the **Tanakpur-Khatima highway and railway line**, ribbon urbanization around Khatima and Chakarpur, and livestock overgrazing (Surai-Khatima-Kilpura Baseline Study). Across Terai Central and Terai West divisions, historical replacement of native Sal and swamp forests with monoculture plantations of exotic **Eucalyptus and Poplar** has reduced understory browse and canopy cavities. Industrial effluents around Kashipur, Bazpur, Kichha, and Sitarganj pollute the Dhela, Baur, and Baigul rivers, while pesticide runoff and illegal gill-net fishing threaten waterfowl and turtle populations in Tumaria, Haripura-Baur, and Nanak Sagar reservoirs.

13. Uttarkashi District (Garhwal Division)

1. Ecosystem Profiles & Geographical Features

- **Protected Areas & Sanctuaries:** Covering 8,016 sq km across the northwestern frontier of Uttarakhand (bordering Himachal Pradesh to the northwest and Tibet/China to the northeast), Uttarkashi district encompasses the largest total protected area expanse of any district in the state—anchoring both the **largest National Park in Uttarakhand** and the **oldest Wildlife Sanctuary in Uttarakhand** (UKPSC Protected Areas):
  - **Gangotri National Park (2,390.02 sq km):** Established in 1989 across the upper catchment of the Bhagirathi River, Jadh Ganga (Nelang Valley), and Kedar Ganga from 1,800 to 7,083 meters (National Parks in Uttarakhand). It forms a continuous high-altitude conservation landscape connecting Govind National Park in the west with Kedarnath Wildlife Sanctuary in the southeast, and encloses the **30-km-long Gangotri Glacier and Gaumukh snout (4,023 m)**—the glacial source of the Ganges (Gomukh Wikipedia).
  - **Govind Pashu Vihar National Park and Wildlife Sanctuary (957.97 sq km total):** Established on **1 March 1955** as Uttarakhand's oldest wildlife sanctuary in the Tons-Supin-Rupin river basin (Purola/Mori subdivision, 1,300–6,387 m at Swargarohini/Bandarpunch), with an inner core of **472.08 sq km** notified as **Govind National Park** in 1990 and **485.89 sq km** retained as **Govind Pashu Vihar Wildlife Sanctuary** (UnlockIAS Protected Area Dates). India's flagship **Project Snow Leopard** and **SECURE Himalaya Project** in Uttarakhand are managed from this landscape.
  - **Bhagirathi Eco-Sensitive Zone (4,179.59 sq km):** Notified by MoEFCC in December 2012 along the 100-km stretch of the Bhagirathi River from **Gaumukh to Uttarkashi town**.
- **Forest Cover & Types:** Managed by five major divisions—Uttarkashi FD, Tons FD (Purola), Upper Yamuna Barkot FD, Gangotri National Park Division, and Govind Pashu Vihar NP & WLS Division (Naitwar/Dharkot)—Uttarkashi holds **3,036.15 sq km of forest cover** (third-largest by area in the state; 37.88% of geographical area, as >60% of the district lies above the 3,600-meter climatic treeline), comprising 566.15 sq km Very Dense, 1,680.00 sq km Moderately Dense,

and 790.00 sq km Open Forest (ISFR District Data). In the lower Tons, Yamuna (Barkot/Naugaon), and Bhagirathi (Chinyalisaur/Dunda) valleys (900–1,800 m), **Asia's largest contiguous pure Chir pine (*Pinus roxburghii*) forest belt** occurs in the Tons and Purola divisions. Between 1,800 and 3,200 meters across Harshil, Mukhba, Dharali, Bhairoghati, Sankri, Taluka, Osla, and Yamunotri, Moist and Dry Temperate Coniferous and Broadleaf Forests of **Himalayan Deodar (*Cedrus deodara*)**, blue pine (*Pinus wallichiana*), West Himalayan spruce (*Picea smithiana*), silver fir (*Abies pindrow*), Banj, Moru, and Kharsu oaks, maple, horse chestnut, walnut, hazel (*Corylus jacquemontii*), and **Chilgoza pine (*Pinus gerardiana*, rare dry-temperate relict near Bhairoghati)** flourish. Above 3,200 meters, sub-alpine **Bhojpatra (*Betula utilis*) forests at Bhojbasa** and *Rhododendron campanulatum* scrub give way to two distinct high-altitude biomes:

  - *Greater Himalayan Alpine Bugyals (3,000–4,500 m):* **Dayara Bugyal** (3,400 m, covering 28 sq km above Raithal/Barsu), **Gidara Bugyal** (4,200 m, largest high-altitude meadow in Uttarakhand), **Har-ki-Dun Bugyal** (3,566 m), **Kedarkantha**, **Dev Kyara**, **Kush Kalyan**, **Tapovan–Nandanvan** (4,463 m above Gaumukh), and **Manji Bugyal**.
  - *Cold-Arid Trans-Himalayan Steppe (3,400–5,500 m):* The **Nelang and Jadhang (Jadh Ganga) Valley** inside Gangotri NP, exhibiting rain-shadow Tibetan plateau vegetation (*Caragana*, *Artemisia*, *Ephedra gerardiana*, and *Juniperus semiglobosa*).

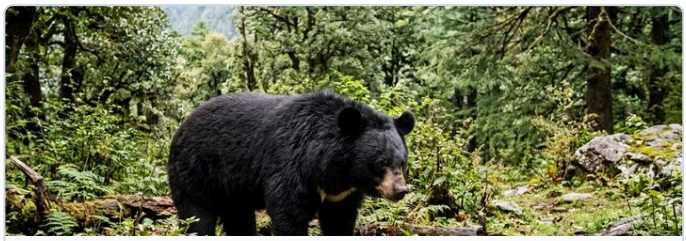
- **Aquatic Ecosystems & Water Bodies:** Headwater source of India's two most sacred rivers: (1) the **Bhagirathi River**, emerging from the **Gaumukh snout of the Gangotri Glacier** and joined by the **Kedar Ganga** (at Gangotri), **Jadh Ganga / Jahnavi** (from Nelang at **Bhairoghati**), **Kakora Gad** and **Jalandhari Gad** (at Harshil), **Pilang Gad**, **Assi Ganga** (originating from Dodital and joining at **Gangori**), and **Varuna River** (at Uttarkashi); and (2) the **Yamuna River**, originating from the **Champasar Glacier above Yamunotri (3,293 m)** and flowing past Janki Chatti, Hanuman Chatti, and Barkot, joined in the west by the **Tons River**—formed at **Naitwar Prayag** by the confluence of the **Rupin River** and **Supin River** (draining Har-ki-Dun and Ruinsara) and receiving the **Kamal River** at Purola. High-altitude lakes include **Dodital** (3,024–3,657 m), **Kedar Tal** (4,750 m below Thalay Sagar), **Ruinsara Tal** (3,500 m in Govind NP), **Bharadsar Tal** (4,400 m), **Nachiketa Tal** (Chaurangi Khal), **Saptarishi Kund**, and **Lama Tal**.

2. Wildlife & Faunal Catalogue

Featured Wildlife & Faunal Species Photograph Gallery — 13. Uttarkashi District:



Mammal: Snow Leopard (Panthera uncia) — Gangotri National Park & Nelang Valley



Mammal: Himalayan Black Bear (Ursus thibetanus laniger) — Govind Pashu Vihar & Harshil



Bird: Himalayan Monal (Lophophorus impejanus) — Dayara Bugyal & Har-ki-Dun alpine timberline



Mammal: Alpine Musk Deer (Moschus chrysogaster) — Sub-alpine birch-rhododendron forests

- **Mammals:** Gangotri National Park (specifically the **Nelang-Jadh Ganga Valley**, Kedar Tal, and Gaumukh-Tapovan) and Govind Pashu Vihar National Park (Har-ki-Dun, Ruinsara, and Borasu Pass) form the **core stronghold of the Snow leopard (*Panthera uncia*) in Uttarakhand**, supporting over half of the state's 124 estimated snow leopards (SPAI Snow Leopard Assessment). In 2017, a snow leopard near Gangotri NP was photographed carrying a freshly killed **Woolly flying squirrel (*Eupetaurus cinereus*)** (Snow Leopard Wikipedia). Other Trans-Himalayan and alpine carnivores include the **Tibetan wolf (*Canis lupus chanco*)**, **Eurasian lynx (*Lynx lynx isabellinus*)**, **Pallas's cat (*Otocolobus manul*, recorded via camera traps in Nelang Valley)**, **Tibetan sand fox (*Vulpes ferrilata*)**, red fox (*Vulpes vulpes montana*), and Himalayan brown bear (*Ursus arctos isabellinus*). Ungulates in Nelang and Gangotri NP include huge herds of **Bharal (*Pseudois nayaur*, numbering several thousand across Nelang, Bhojbasa, and Tapovan)**, **Tibetan argali (*Ovis ammon hodgsoni*)**, **Tibetan gazelle (*Procapra picticaudata*)**, **Alpine musk deer**, and **Himalayan tahr**. Temperate forests of Govind Pashu Vihar, Dodital, Harshil, and Barkot support the Indian leopard,

Asiatic black bear, Himalayan serow, Himalayan goral, sambar, barking deer, leopard cat, yellow-throated marten, stone marten (*Martes foina*), Himalayan marmot (*Marmota himalayana* in Nelang), and Royle's pika.

- **Birds (Avifauna):** Govind Pashu Vihar National Park (Supin-Har-ki-Dun-Kedarkantha) and **Dodital–Dayara Bugyal** are critical Important Bird Areas (IBAs) supporting over **280 avian species**. Govind Pashu Vihar is the **easternmost viable stronghold of the globally Vulnerable Western tragopan (*Tragopan melanocephalus*)**, alongside Satyr tragopan, State Bird Himalayan monal (*Lophophorus impejanus*), Cheer pheasant (*Catreus wallichii*, in the Sankri-Naitwar-Tons grassy cliffs), Koklass pheasant, Kalij pheasant, Tibetan snowcock (*Tetraogallus tibetanus* in Nelang), Himalayan snowcock, and Tibetan sandgrouse (*Syrrhaptes tibetanus*). Raptors include the Bearded vulture, golden eagle, steppe eagle (*Aquila nipalensis*), and Himalayan griffon, alongside orange bullfinch, spectacled finch, white-browed tit-warbler, grandala, and wallcreeper.
- **Reptiles & Amphibians:** Includes the Himalayan pit viper (*Gloydius himalayanus*, recorded up to 3,500 m in Govind NP and Dodital), Himalayan keelback (*Herpetoreas platyceps*), common krait and rat snake (lower Barkot/Chinyalisaur valleys), Himalayan rock agama (*Paralaudakia himalayana*—abundant on granite boulders from Harshil to Gangotri), Kashmir rock agama (*Laudakia tuberculata*), Sikimmese skink, and high-altitude amphibians (*Duttaphrynus himalayanus*, *Nanorana liebigii*, and *Amolops* spp.).
- **Insects & Butterflies:** Har-ki-Dun, Ruinsara, Dayara, and Gangotri alpine valleys support high-altitude Apollo butterflies —**Common Red Apollo (*Parnassius epaphus*)**, **Common Blue Apollo (*Parnassius hardwickii*)**, **Keeled Apollo (*Parnassius jacquemontii*)**, and **Regal Apollo (*Parnassius charltonius*)**—alongside Clouded Yellows (*Colias*), Mountain Argus (*Aricia artaxerxes*), and Common Peacock (*Papilio bianor* in Tons/Dodital).
- **Aquatic Species:** **Dodital** and the **Assi Ganga River** are renowned for their populations of **Himalayan Brown Trout (*Salmo trutta fario*, introduced in the early 20th century)** and native **Snow Trout (*Schizothorax richardsonii* and *Schizothorax plagiostomus*)**. In the **Tons, Rupin, Supin, and Yamuna (Barkot)** rivers, snow trout, Golden mahseer (*Tor putitora*, in lower Yamuna-Tons reaches near Mori/Tiuni), and torrent catfishes (*Glyptothorax* and *Pseudecheneis*) thrive.

3. Habitat, Distribution, and Conservation Status

- **Micro-Habitats & Altitudinal Migration:** The **Nelang–Gangotri–Govind continuum** represents the largest unbroken high-altitude wilderness in Uttarakhand (>3,348 sq km combined). Within this complex, the cold-arid rain-shadow cliffs and steppes of the **Nelang Valley** (3,400–5,200 m) provide snow-free winter grazing for massive herds of Bharal and Tibetan argali, sustaining the highest breeding density of Snow leopard in the state. Meanwhile, the moist *Quercus semecarpifolia–Abies pindrow–Ringal* forests of the **Supin and Rupin valleys in Govind Pashu Vihar National Park** (2,400–3,300 m) safeguard the easternmost population of the Western tragopan.
- **Statutory Conservation Status:** Priority Schedule I species include the Snow leopard (IUCN Vulnerable; CITES Appendix I), Alpine musk deer (IUCN Endangered; CITES Appendix I), Woolly flying squirrel (IUCN Endangered), Pallas's cat (IUCN Least Concern; CITES Appendix II), Tibetan argali (IUCN Near Threatened; CITES Appendix I), Tibetan gazelle (IUCN Near Threatened; CITES Appendix I), Western tragopan (IUCN Vulnerable; CITES Appendix I), and Cheer pheasant (IUCN Vulnerable; CITES Appendix I).

4. Human-Wildlife Conflict & Environmental Threats

- **Conflict Hotspots:** In **Govind Pashu Vihar WLS (Mori, Sankri, Osla, Jakhol, Naitwar)** and **Bhatwari/Dunda/Naugaon/Purola**, the primary conflict species are the **Asiatic black bear (*Ursus thibetanus*)** and **Indian leopard**. Black bears raid red rice, amaranth (*Chaulai*), buckwheat (*Ogal/Phaphar*), maize, and high-value apple and walnut orchards in Harshil, Arakot, Mori, and Sankri, frequently mauling farmers in autumn. Leopards prey on cattle, mules, and goats around Uttarkashi, Chinyalisaur, Barkot, and Purola, while snow leopards and Tibetan wolves occasionally depredate sheep-goat herds of *Gaddi* and local pastoralists in the upper Supin, Rupin, and Harshil valleys.
- **Environmental Threats:** Uttarkashi's fragile ecosystems face three major cumulative threats: (1) **Glacier Recession & Cryospheric Shifts:** The **Gangotri Glacier (30.2 km long)** is retreating at **15 to 22 meters per year**, accompanied by an upward shift of the *Betula utilis* and *Pinus wallichiana* treeline by ~10–15 meters per decade, compressing alpine nival habitats for snow leopards and pikas; (2) **Char Dham Highway Widening & Tunneling:** Widening of the **Bhagirathi Eco-Sensitive Zone highway (NH-34 to Gangotri)**—threatening mature **Harshil-Bhairoghati Himalayan Deodar (*Cedrus deodara*)** stands—and tunneling along the **Yamunotri Highway (NH-134, site of the November 2023 Silkyara-Barkot tunnel collapse)** (Silkyara Tunnel Collapse Wikipedia) have triggered slope instability and muck dumping (Char Dham Environmental Crisis); and (3) **Hydropower Dams & Mass Trekking:** Operational dams (**Maneri Bhali Stage I & II** on the Bhagirathi and **Netwar-Mori** on the Tons) fragment lotic habitats, while heavy winter/spring trekking to **Kedarkantha, Har-ki-Dun, and Dayara Bugyal** causes alpine turf erosion (mitigated at **Dayara Bugyal** using biodegradable Pirul pine-needle and jute geo-coir mats across 28 sq km) (Dayara Bugyal Restoration).

Master Taxonomic & Statutory Conservation Matrix (IUCN, CITES & WPA 1972)

Common Name	Scientific Name	Taxonomic Group	Altitudinal Range in Uttarakhand (m)	IUCN Red List Status	CITES Appendix	WPA (1972 / 2022 Amend.)	Primary District Strongholds
Bengal tiger	*Panthera tigris tigris*	Mammalia	190 – 3,630 m	Endangered (EN)	Appendix I	Schedule I	Nainital, Pauri, Haridwar, Dehradun, US Nagar, Champawat, Pithoragarh, Rudraprayag
Snow leopard	*Panthera uncia*	Mammalia	3,000 – 5,800 m	Vulnerable (VU)	Appendix I	Schedule I	Uttarkashi, Chamoli, Pithoragarh, Bageshwar, Rudraprayag, Tehri Garhwal
Indian leopard	*Panthera pardus fusca*	Mammalia	190 – 3,400 m	Vulnerable (VU)	Appendix I	Schedule I	All 13 Districts (peak conflict in Pauri, Almora, Tehri, Rudraprayag, Pithoragarh)
Indian elephant	*Elephas maximus indicus*	Mammalia	190 – 1,500 m	Endangered (EN)	Appendix I	Schedule I	Haridwar, Dehradun, Pauri, Nainital, Champawat, Udham Singh Nagar
Alpine musk deer (State Animal)	*Moschus chrysogaster / M. leucogaster*	Mammalia	2,500 – 4,300 m	Endangered (EN)	Appendix I	Schedule I	Chamoli, Rudraprayag, Pithoragarh, Uttarkashi, Bageshwar, Tehri Garhwal
Himalayan tahr	*Hemitragus jemlahicus*	Mammalia	2,200 – 4,400 m	Near Threatened (NT)	Not Listed	Schedule I	Chamoli, Rudraprayag, Bageshwar, Uttarkashi, Pithoragarh, Tehri Garhwal
Himalayan serow	*Capricornis sumatraensis thar*	Mammalia	1,800 – 3,300 m	Vulnerable (VU)	Appendix I	Schedule I	Chamoli, Rudraprayag, Pithoragarh, Bageshwar, Uttarkashi, Nainital, Almora, Tehri
Bharal (Himalayan Blue Sheep)	*Pseudois nayaur*	Mammalia	3,200 – 5,500 m	Least Concern (LC)	Appendix III	Schedule I	Uttarkashi (Nelang/Gangotri), Chamoli (Nanda Devi), Pithoragarh, Bageshwar
Himalayan goral	*Naemorhedus goral*	Mammalia	600 – 3,000 m	Near Threatened (NT)	Appendix I	Schedule I	All 11 Hill & Shivalik Districts (Almora, Nainital, Pauri, Tehri, Dehradun, etc.)
Swamp deer (Barasingha)	*Rucervus duvaucelii*	Mammalia	190 – 300 m	Vulnerable (VU)	Appendix I	Schedule I	Haridwar (Jhilmil Jheel, Baan Ganga), Udham Singh Nagar (Terai East/Central)
Hog deer	*Axis porcinus*	Mammalia	190 – 600 m	Endangered (EN)	Appendix I	Schedule I	Pauri Garhwal (Dhikala Chaur), Nainital, Haridwar (Jhilmil), Udham Singh Nagar
Asiatic black bear	*Ursus thibetanus*	Mammalia	1,200 – 3,600 m	Vulnerable (VU)	Appendix I	Schedule I	Chamoli, Uttarkashi, Rudraprayag, Pithoragarh, Bageshwar, Tehri, Almora, Nainital
Himalayan brown bear	*Ursus arctos isabellinus*	Mammalia	3,200 – 5,000 m	Least Concern (CR in India)	Appendix I	Schedule I	Uttarkashi (Gangotri/Govind), Chamoli (Nanda Devi/Valley of Flowers), Pithoragarh
Sloth bear	*Melursus ursinus*	Mammalia	190 – 1,000 m	Vulnerable (VU)	Appendix I	Schedule I	Nainital, Pauri, Haridwar, Dehradun, Champawat, Udham Singh Nagar
Smooth-coated otter	*Lutrogale perspicillata*	Mammalia	190 – 1,400 m	Vulnerable (VU)	Appendix I	Schedule I	Pauri (Ramganga/Palain), Nainital (Kosi), Haridwar, US Nagar, Champawat
Indian pangolin	*Manis crassicaudata*	Mammalia	200 – 1,600 m	Endangered (EN)	Appendix I	Schedule I	Dehradun, Haridwar, Pauri Garhwal, Nainital, Almora, Champawat
Woolly flying squirrel	*Eupetaurus cinereus*	Mammalia	2,400 – 3,800 m	Endangered (EN)	Not Listed	Schedule I	Uttarkashi (Gangotri NP), Chamoli (Dhauliganga/Niti), Pithoragarh

Common Name	Scientific Name	Taxonomic Group	Altitudinal Range in Uttarakhand (m)	IUCN Red List Status	CITES Appendix	WPA (1972 / 2022 Amend.)	Primary District Strongholds
Himalayan monal (State Bird)	*Lophophorus impejanus*	Aves	2,100 – 4,500 m	Least Concern (LC)	Appendix I	Schedule I	Rudraprayag, Chamoli, Uttarkashi, Pithoragarh, Bageshwar, Tehri, Almora
Western tragopan	*Tragopan melanocephalus*	Aves	2,400 – 3,600 m	Vulnerable (VU)	Appendix I	Schedule I	Uttarkashi (Govind Pashu Vihar), Bageshwar (Upper Pindar), Chamoli
Cheer pheasant	*Catreus wallichii*	Aves	1,400 – 3,000 m	Vulnerable (VU)	Appendix I	Schedule I	Nainital (Pangot), Dehradun (Benog/Chakrata), Uttarkashi, Tehri, Almora, Pithoragarh
Bearded vulture (Lammergeier)	*Gypaetus barbatus*	Aves	1,500 – 5,500 m	Near Threatened (NT)	Appendix II	Schedule I	Chamoli, Uttarkashi, Rudraprayag, Pithoragarh, Bageshwar, Dehradun
Red-headed vulture	*Sarcogyps calvus*	Aves	200 – 2,000 m	Critically Endangered (CR)	Appendix II	Schedule I	Dehradun (Asan), Nainital (Corbett), Pauri Garhwal (Dhikala), Haridwar
Finn's weaver	*Ploceus megarhynchus*	Aves	190 – 260 m	Endangered (EN)	Not Listed	Schedule I	Udham Singh Nagar (Haripura-Baur, Tumaria), Haridwar (Jhilmil Jheel)
Great hornbill	*Buceros bicornis*	Aves	200 – 1,200 m	Vulnerable (VU)	Appendix I	Schedule I	Nainital (Corbett/Pawalgarh/Nandhaur), Pauri, Dehradun, Haridwar, Champawat
Gharial	*Gavialis gangeticus*	Reptilia	190 – 500 m	Critically Endangered (CR)	Appendix I	Schedule I	Pauri Garhwal & Nainital (Ramganga & Kosi in Corbett), Haridwar (Ganges)
Mugger crocodile	*Crocodylus palustris*	Reptilia	190 – 600 m	Vulnerable (VU)	Appendix I	Schedule I	US Nagar (Kakra Nala, Baur, Nanak Sagar), Pauri, Nainital, Haridwar, Champawat
King cobra	*Ophiophagus hannah*	Reptilia	200 – 2,200 m	Vulnerable (VU)	Appendix II	Schedule I (2022)	Nainital (Kaladhungi/Chorgaliya), Champawat, Dehradun, Pauri, Haridwar, Pithoragarh
Elongated tortoise	*Indotestudo elongata*	Reptilia	200 – 900 m	Critically Endangered (CR)	Appendix II	Schedule I	Nainital & Pauri Garhwal (Corbett Sal forests), Dehradun & Haridwar (Rajaji)
Golden mahseer (State Fish)	*Tor putitora*	Actinopterygii	190 – 1,800 m	Endangered (EN)	Not Listed	Protected	Champawat & Pithoragarh (Kali/Saryu), Pauri (Ramganga/Nayar), Nainital, Dehradun
Kaiser-i-Hind	*Teinopalpus imperialis*	Insecta (Lepidoptera)	1,800 – 3,000 m	Near Threatened (NT)	Appendix II	Schedule I	Pithoragarh (Munsiyari, Didihat, Askot), Bageshwar
Common Peacock (State Butterfly)	*Papilio bianor polyctor*	Insecta (Lepidoptera)	800 – 2,600 m	Least Concern (LC)	Not Listed	Protected	All 11 Hill & Bhabar Districts (abundant in Tehri/Devalsari, Nainital, Almora, Chamoli)

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- **Dedicated Protected Area & Wetland Sources:** Jim Corbett National Park | Rajaji National Park | Nanda Devi National Park | Valley of Flowers National Park | Gangotri National Park | Govind Pashu Vihar National Park | Kedarnath Wildlife Sanctuary | Askot Musk Deer Sanctuary | Nandhaur Wildlife Sanctuary Official Portal | Binsar Wildlife Sanctuary | Mussoorie / Benog Wildlife Sanctuary | Naina Devi Himalayan Bird Conservation Reserve | Pawalgarh / Sitabani Conservation Reserve | Jhilmil Jheel Conservation Reserve Official Portal | Asan Barrage / Ramsar Site | Upper Pindar Catchment Key Biodiversity Area (KBA 18452)
- **District Portals & Regional Ecological Studies:** Almora District Portal | Bageshwar District Portal | Chamoli District | Champawat District Portal | Dehradun District | Haridwar District Portal | Nainital District | Pauri Garhwal Official Wildlife Portal | Pithoragarh Official Flora & Fauna Portal | Rudraprayag District Portal | Tehri Garhwal Official Fauna Portal | Udham Singh Nagar District | Uttarkashi District
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